

VERIQTA

Production
Engineering
Notebook

KUBERNETES SECURITY HANDBOOK



Enterprise Kubernetes Security for Production Clusters,
Zero Trust, Compliance, and Cloud Native Defense

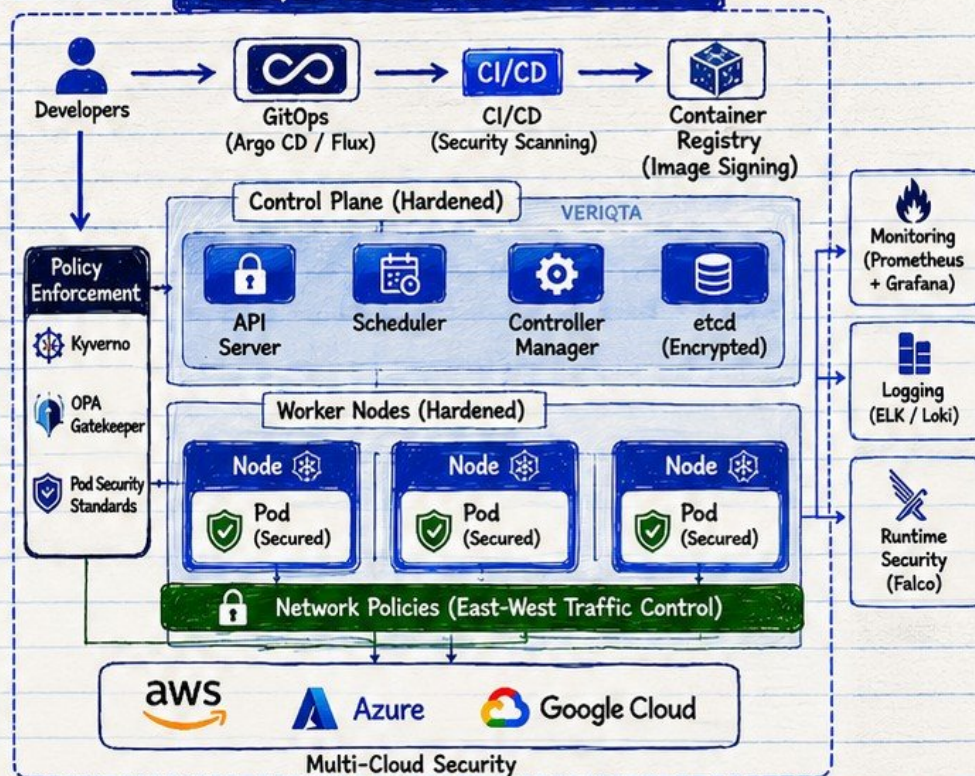
What You'll Master

- ✓ RBAC & IAM
- ✓ Pod & Container Security
- ✓ Network Security
- ✓ Secrets Management
- ✓ Supply Chain Security
- ✓ Policy as Code
- ✓ Runtime Security
- ✓ Cloud Kubernetes
- ✓ GitOps Security
- ✓ Compliance & Governance
- ✓ Incident Response
- ✓ Production Hardening

Perfect For

- DevOps Engineers
- Platform Engineers
- Cloud Engineers
- SRE Engineers
- Security Engineers
- Technical Architects

VERIQTA Secure Kubernetes Cluster



Zero Trust
Architecture

Defense
in Depth

Least Privilege
Access

Continuous
Monitoring

Production
Ready

BUILD. SECURE. OPERATE. PROTECT.

SECURE BY DESIGN. DEFEND IN PRODUCTION.



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Engineering Library
2026 Edition

1. INTRODUCTION TO KUBERNETES SECURITY



WHY KUBERNETES SECURITY MATTERS

- Kubernetes runs critical workloads, sensitive data, and business logic.
- Misconfigurations, weak access controls, and vulnerabilities can lead to **full cluster compromise**.
- Attackers target Kubernetes because it is complex, powerful, and often misused.
- Security must be **built in from the design phase**, not added later.

VERIQTA PRODUCTION NOTE



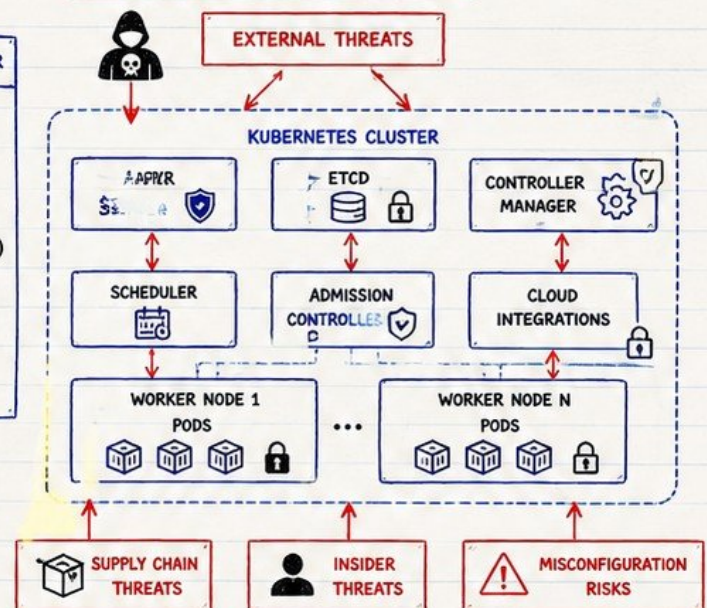
Most Kubernetes breaches are not caused by Kubernetes bugs, but by human mistakes:

- ✓ Over-permissive RBAC
- ✓ Exposed dashboards
- ✓ Weak network policies
- ✓ Secrets in plaintext
- ✓ Untrusted container images

SHARED RESPONSIBILITY MODEL

YOU (CUSTOMER / CLUSTER OPERATOR)	CLOUD / MANAGED SERVICE PROVIDER
• Secure workloads & applications • RBAC & access management • Network policies • Secrets management • Monitoring, logging & alerting • Patching nodes and workloads • Admission policies • Backup & disaster recovery • Compliance & governance	• Control plane availability • etcd management • Underlying infrastructure • Cluster upgrades (control plane) • Managed node groups (OS updates) • Load balancing • Cloud network fabric • Physical security • Service integrations

ATTACK SURFACE OVERVIEW



HANDBOOK OBJECTIVES



- Build a strong security foundation for Kubernetes.
- Understand risks, threats, and mitigation strategies.
- Learn production-grade security controls and best practices.
- Protect workloads, data, and infrastructure end to end.
- Prepare for **real-world attacks**, compliance, and audits.

SECURITY MINDSET



- Think like an **attacker**, defend like an **engineer**.
- Assume breach, **minimize blast radius**.
- Enforce **least privilege** everywhere.
- **Automate** security, reduce human error.
- Continuously **monitor, detect, and respond**.
- Security is a **continuous process**, not a one-time task.



VERIQTA SENIOR ENGINEER TIP

Secure the foundation first: identity, access, network, and workload boundaries. Everything else (monitoring, scanning, compliance) becomes effective only after the foundation is strong.



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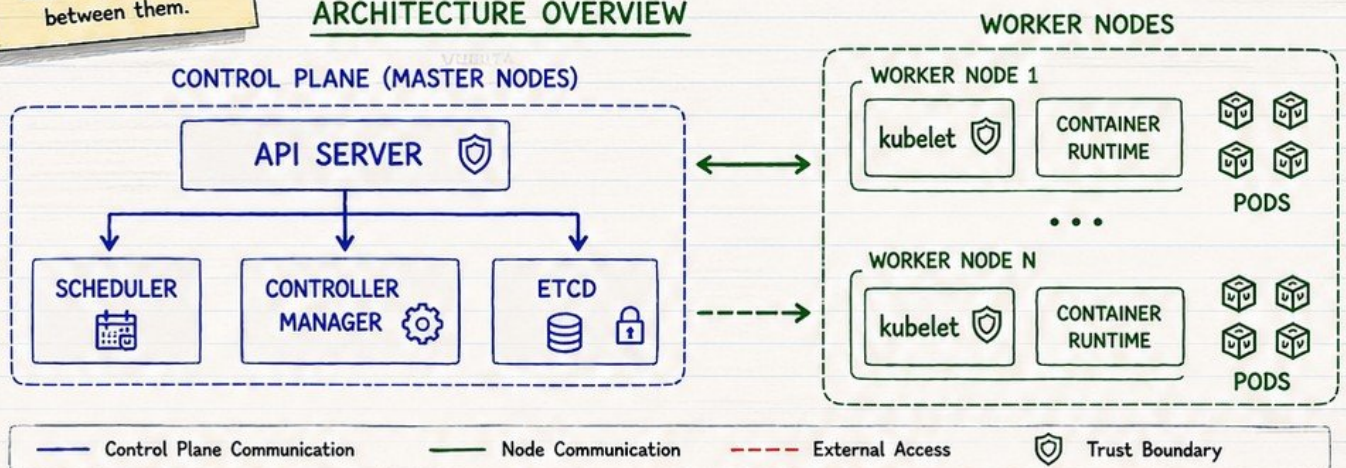
2. KUBERNETES ARCHITECTURE FROM A SECURITY PERSPECTIVE



VERIQTA TIP

Understand the architecture deeply. Security is about protecting components and the boundaries between them.

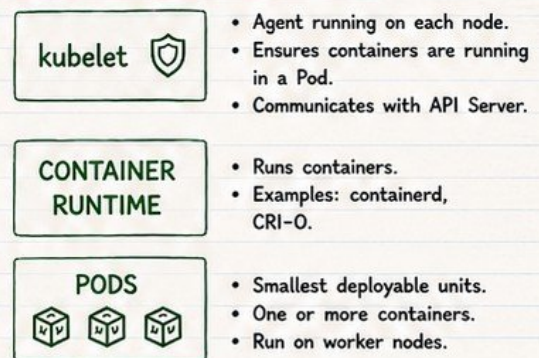
ARCHITECTURE OVERVIEW



CONTROL PLANE COMPONENTS



WORKER NODE COMPONENTS



TRUST BOUNDARIES



Attackers can try to cross these boundaries. Our job is to enforce controls at every boundary.



Strong security = Strong boundaries + Least privilege + Continuous monitoring



VERIQTA PRODUCTION NOTE

If an attacker gains access to etcd or the API Server with admin privileges, the entire cluster is compromised. Protect these components with everything you have.

KEY SECURITY PRINCIPLES

- ✓ Minimize attack surface.
- ✓ Enforce least privilege.
- ✓ Encrypt in transit and at rest.
- ✓ Monitor everything.
- ✓ Assume breach, always verify.



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VERIQTA TIP

Attackers follow paths of least resistance.
Secure every layer.
Assume breach.
Verify everything.

3. KUBERNETES THREAT MODEL



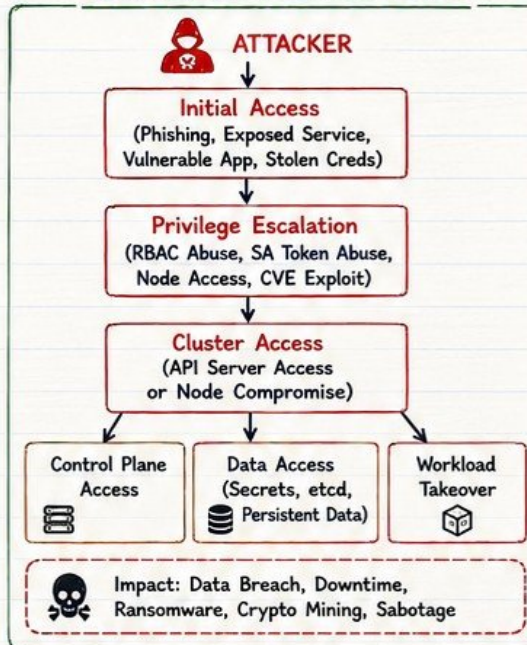
UNDERSTAND THE ENEMY. PROTECT THE CLUSTER.

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COMMON ATTACK VECTORS

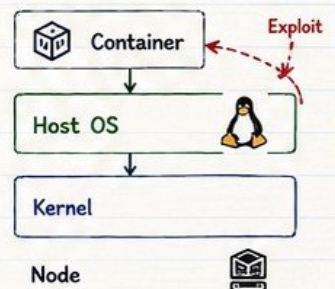
- Exposed Services**
Exposed APIs, dashboards, ingress misconfigurations.
- Weak Identity & Access**
Excessive permissions, weak RBAC, stolen tokens.
- Vulnerable Workloads**
Unpatched images, vulnerable applications, misconfigurations.
- Secrets Exposure**
Secrets in code, plain text, or overly permissive access.
- Network Weakness**
No network policies, excessive pod communication.
- Misconfigurations**
Insecure defaults, unnecessary privileges, open ports.

CLUSTER COMPROMISE PATHS



CONTAINER ESCAPE

- Exploits in kernel, runtime, or misconfigurations can break out of the container.
- Leads to node compromise and full cluster risk.

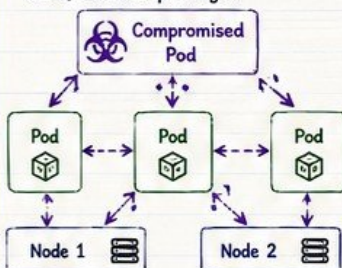


VERIQTA TIP

Use least privilege, seccomp, AppArmor/SELinux, and keep everything patched.

LATERAL MOVEMENT

- After initial access, attackers move between pods and nodes.
- Goal: expand access, find sensitive data, escalate privileges.

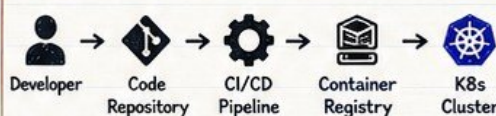


VERIQTA TIP

Implement Network Policies, micro-segmentation, and least privilege communication.

SUPPLY CHAIN ATTACKS

- Attackers compromise the supply chain to inject malicious code.
- Impacts the entire delivery pipeline.



RISKS

- Malicious code injection
- Compromised base images
- Dependency confusion
- Signed image bypass
- Unverified artifacts

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Use image signing, SBOM, vulnerability scanning, trusted registries, and pipeline security.

INSIDER THREATS

- Insiders can intentionally or accidentally cause harm.
- Often bypass perimeter defenses.



- Excessive Permissions
- Data Exfiltration
- Secret Access
- Malicious Deployments
- Sabotage

VERIQTA TIP

Enforce least privilege, audit everything, monitor behavior, and separate duties.

KEY TAKEAWAY



Kubernetes security is not one control. It is a combination of strong identity, secure workloads, hardened nodes, network isolation, observability, and continuous validation.



ATTACKERS TARGET:

- Exposed Services
- Weak Access
- Vulnerable Workloads
- Secrets
- Misconfigurations
- Unrestricted Networks



YOUR DEFENSE:

- Least Privilege
- Zero Trust
- Defense in Depth
- Continuous Monitoring
- Automate Security
- Assume Breach

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SENIOR ENGINEER NOTE

Security is not a product, it is a process.
Build it in.
Test it often.
Trust it never.



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4. AUTHENTICATION



VERIQTA TIP

Strong authentication is the first line of defense for your Kubernetes cluster. Secure identities = Secure cluster.

USER AUTHENTICATION



- Human users authenticate to the Kubernetes API Server to perform actions.
- Use strong, centralized identity providers.
- Enforce MFA for all users.
- Use short-lived tokens.

SERVICE ACCOUNTS



- Used by pods and controllers to authenticate to the API.
- Each Service Account has a unique token.
- Bind to least privilege RBAC roles.
- Automount only when necessary.

CERTIFICATES



- TLS certs secure all communication.
- API Server, etcd, kubelet, and admin access use certificates.
- Use strong CA and rotate regularly.
- Verify cert SANs and expiry.

OIDC (OPENID CONNECT)



- Modern authentication using OpenID Connect.
- Integrates with external Identity Providers (IdP).
- Supports SSO and MFA.
- Issues short-lived tokens (JWT).

IDENTITY PROVIDERS (IdP)



Azure AD



Google Workspace



GitHub



Okta



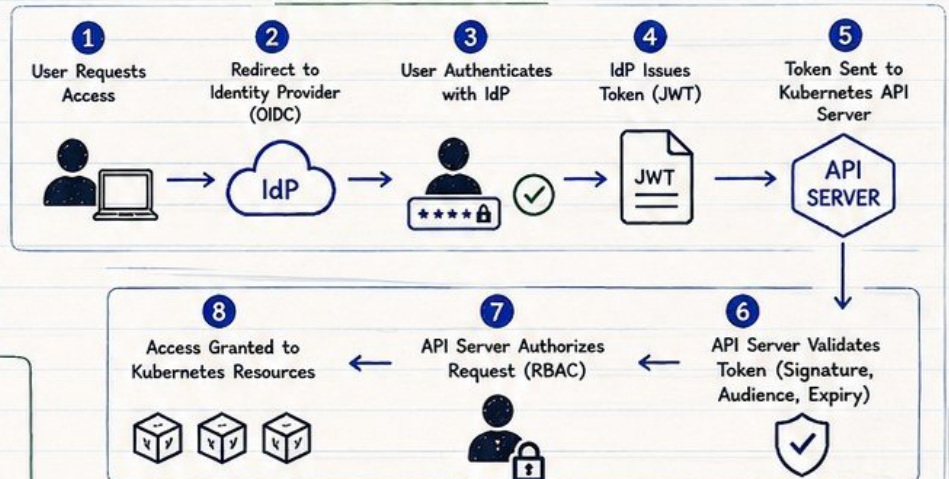
Keycloak

- Centralized identity management.
- Supports SAML, OIDC, and LDAP.
- Enforce MFA, conditional access, and policies.
- Map groups/roles to Kubernetes RBAC.

BEST PRACTICES

- ✓ Enable MFA for all human users.
- ✓ Use OIDC with short-lived tokens.
- ✓ Disable anonymous access.
- ✓ Use least privilege RBAC.
- ✓ Do not use long-lived Service Account tokens.
- ✓ Rotate certificates before expiry.
- ✓ Use strong CA and protect private keys.
- ✓ Audit authentication logs regularly.
- ✓ Restrict access by IP, network, and context.
- ✓ Apply Zero Trust: verify every request.

AUTHENTICATION FLOW



TOKEN TYPES

TOKEN TYPE	USED BY	DURATION	STORAGE	NOTES
Service Account	Pods, Controllers	Long-lived	Secret (K8s)	Avoid automount unless needed
OIDC (JWT)	Users, Service Accounts (via IdP)	Short-lived	In-memory	Recommended for users
Bootstrap Token	Nodes joining the cluster	Short-lived	Secret (K8s)	Used during node registration
Client Certificate	Users, Admins, Components	Long-lived	File / Disk	Use cert rotation and strong CA



COMMON AUTH RISKS

- Using default Service Accounts.
- Excessive permissions in RBAC.
- Long-lived tokens and certificates.
- No MFA or weak IdP policies.
- Exposed or stolen credentials.



VERIQTA PRODUCTION NOTE

Authentication is the foundation of Kubernetes security. If an attacker can authenticate, everything else can be compromised. Protect identities, enforce strong policies, and monitor authentication continuously.



CHECKLIST

- ✓ OIDC configured
- ✓ MFA enforced
- ✓ RBAC least privilege
- ✓ Service Accounts reviewed
- ✓ Certs valid and rotated
- ✓ Audit logs enabled

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VERIQTA TIP

Authorization answers
"What can this
identity do?"

Not who they are.
That's authentication.

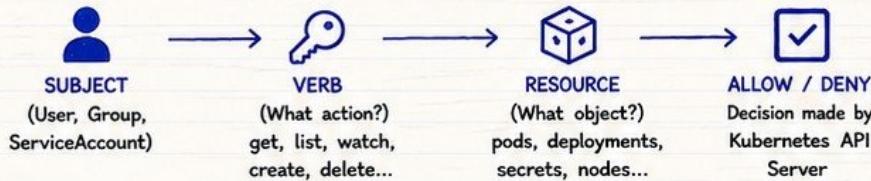
5. AUTHORIZATION

CONTROL ACCESS. ENFORCE LEAST PRIVILEGE.



RBAC FUNDAMENTALS

Kubernetes uses Role-Based Access Control (RBAC) to authorize users and workloads.



LEAST PRIVILEGE PRINCIPLE



Grant only the minimum permissions required to perform a task.

- ✓ Reduce blast radius
- ✓ Prevent privilege escalation
- ✓ Improve security posture
- ✓ Simplify compliance

ROLES

Define permissions within a specific namespace.



Example:
kind: Role
metadata:
 name: pod-reader
 namespace: dev
rules:
- apiGroups: [""]
 resources: ["pods"]
 verbs: ["get", "list", "watch"]

CLUSTERROLES

Define permissions cluster-wide (not limited to a namespace).



Example:
kind: ClusterRole
metadata:
 name: cluster-admin
rules:
- apiGroups: ["*"]
 resources: ["*"]
 verbs: ["*"]

ROLEBINDINGS

Grant a Role to subjects within a namespace.



Example:
kind: RoleBinding
metadata:
 name: read-pods
 namespace: dev
subjects:
- kind: User
 name: alice
roleRef:
 kind: Role
 name: pod-reader
apiGroup: rbac.authorization.k8s.io

CLUSTERROLEBINDINGS

Grant a ClusterRole to subjects cluster-wide.



Example:
kind: ClusterRoleBinding
metadata:
 name: admin-binding
subjects:
- kind: Group
 name: platform-team
roleRef:
 kind: ClusterRole
 name: cluster-admin
apiGroup: rbac.authorization.k8s.io

SUBJECTS

Can be Users, Groups, or Service Accounts.



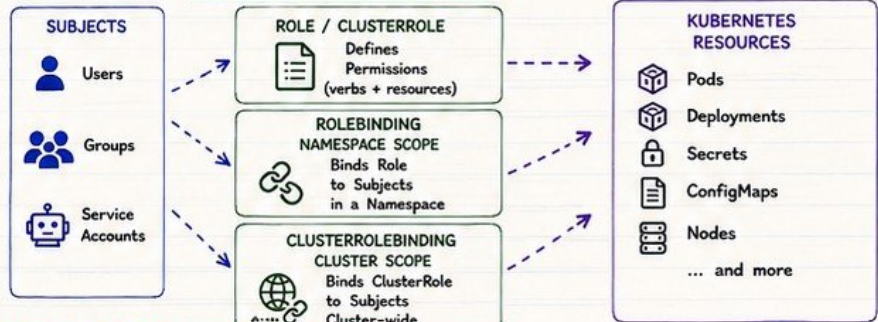
Examples:

- User: alice
- Group: devs
- ServiceAccount: build-bot (in CI/CD namespace)

HOW RBAC WORKS

1. A subject (user, group, or service account) makes a request to the API Server.
2. API Server authenticates the request.
3. API Server checks authorization using RBAC (roles, bindings, permissions).
4. If allowed → request proceeds.
If denied → access is blocked.

RBAC INTERACTION DIAGRAM



COMMON RBAC MISTAKES

OVERLY PERMISSIVE ROLES

Using cluster-admin or * verbs/resources unnecessarily.



BINDING TO EVERYONE

Binding roles to system:authenticated or large groups carelessly.



USING CLUSTER SCOPE UNNECESSARILY

Using ClusterRoles and ClusterRoleBindings when namespace-scoped access is enough.



FORGETTING TO REVIEW SERVICE ACCOUNTS

Default service accounts may have more access than required.



NOT AUDITING RBAC REGULARLY

Permissions creep leads to privilege sprawl.



BEST PRACTICES

- ✓ Follow least privilege always.
- ✓ Use namespace-scoped Roles when possible.
- ✓ Review and audit RBAC regularly.
- ✓ Bind roles to dedicated service accounts.
- ✓ Avoid using cluster-admin.
- ✓ Document and version control RBAC manifests.
- ✓ Use tools like rbac-tool and kubectrl auth can-i.



VERIQTA PRODUCTION NOTE



Strong authorization is your last line of defense after authentication. Proper RBAC reduces risk, limits damage, and ensures your cluster stays secure and compliant.



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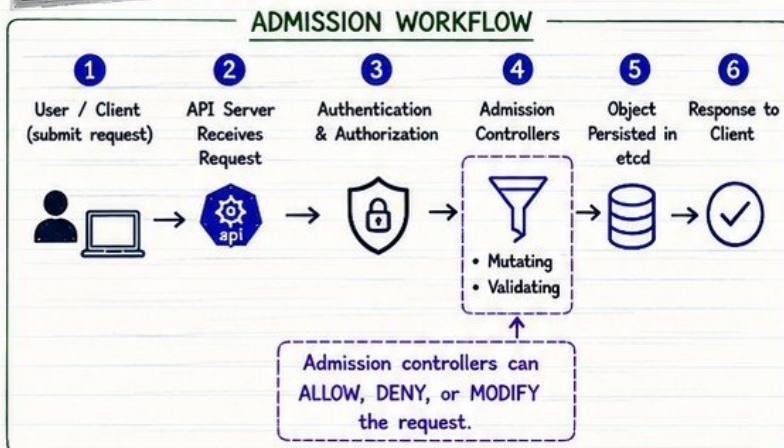
Admission controllers are your cluster gatekeepers. They intercept requests before objects are saved to etcd.

6. ADMISSION CONTROLLERS

CONTROL WHAT ENTERS YOUR CLUSTER.

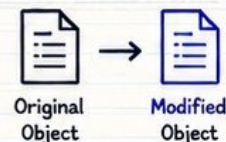


TYPES OF ADMISSION



MUTATING ADMISSION

- Can modify the object before it is stored.
- Common use: defaulting, adding labels, sidecars, annotations.



VALIDATING ADMISSION

- Can allow or deny the request.
- Cannot modify the object.
- Common use: enforce policies, security rules, compliance.



BUILT-IN ADMISSION CONTROLLERS

CONTROLLER	PURPOSE
NamespaceLifecycle	Prevent operations on terminating namespaces.
LimitRanger	Enforce resource limits and requests.
ServiceAccount	Automatically add default ServiceAccount.
DefaultStorageClass	Set default StorageClass for PVCs.
ResourceQuota	Enforce resource quotas in namespaces.
NodeRestriction	Restrict kubelet from modifying nodes.
PodSecurity	Enforce Pod Security Standards (baseline/restricted).
ImagePolicyWebhook	Validate container images via webhook.
AlwaysPullImages	Ensure images are always pulled.
DenyExecOnPrivileged	Deny exec into privileged containers.
EventRateLimit	Prevent event flooding.
... and more	Extensible via webhooks.

POLICY ENFORCEMENT

HOW IT WORKS

- Requests pass through admission chain.
- Mutating controllers can change the object.
- Validating controllers evaluate the final object.
- If any controller denies, the request is rejected.

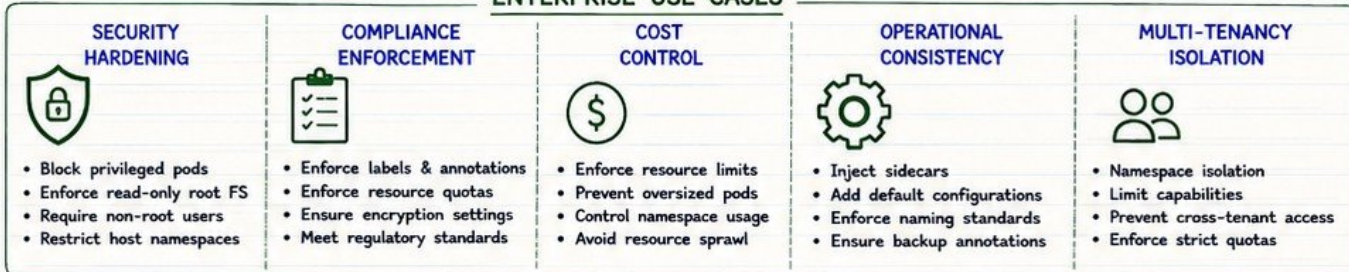
COMMON TOOLS

- Kyverno (Policy as Code)
- OPA Gatekeeper (Rego policies)
- Kubernetes Validating Admission Webhook
- Pod Security Standards (PSS)
- Open Policy Agent (OPA)

POLICY EXAMPLES

- ✓ Only allow images from approved registries
- ✓ Enforce resource limits on all containers
- ✓ Block privileged containers
- ✓ Require labels on all resources
- ✓ Enforce non-root users

ENTERPRISE USE CASES



COMMON CHALLENGES

- Misconfigured policies can block critical workloads.
- Policy order and dependencies matter.
- Webhook downtime can impact cluster operations.
- Overly strict policies can slow down delivery.



VERIQTA PRODUCTION NOTE

Admission controllers are essential for enterprise Kubernetes security. Start with built-in controllers, then layer policies using Kyverno or OPA Gatekeeper.



KEY TAKEAWAY

You can't secure what you don't control. Admission controllers let you enforce rules before anything enters your cluster.



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7. POD SECURITY STANDARDS



VERIQTA TIP

Pod Security Standards replace Pod Security Policies (PSP). Simple, scalable, and built-in to Kubernetes.

SECURE YOUR PODS. PROTECT YOUR CLUSTER.

LEVEL COMPARISON (SUMMARY)

CONTROL	PRIVILEGED	BASILINE	RESTRICTED
Host PID/IPC/Network	✓	✗	✗
Privileged Containers	✓	✗	✗
HostPath Volumes	✓	Limited	✗
Allow Privilege Escalation	✓	✗	✗
Run as Root	✓	Allowed	✗
Capabilities (add)	All	Limited	✗
Seccomp	Unconfined	Runtime Default	Runtime Default
AppArmor/SELinux	Unconfined	Optional	Enforced

POD SECURITY STANDARD LEVELS

PRIVILEGED



- Unrestricted policy.
- Allows known privilege escalations.

- Not recommended for general use.

Use case: Legacy systems, trusted workloads only.

BASILINE



- Minimizes known privilege escalations.
- Blocks privilege escalation and host namespace access.
- Allows common needs.

Use case: General workloads with moderate isolation.

RESTRICTED



- Highly restrictive policy.
- Follows security best practices.
- Strong isolation by default.

Use case: Multi-tenant or high security workloads.

NAMESPACE ENFORCEMENT

Pod Security Standards are enforced at the namespace level using labels.

```
# Label a namespace with a level
kubectl label namespace <ns-name> \
  pod-security.kubernetes.io/enforce=baseline \
  pod-security.kubernetes.io/enforce-version=latest
```

LABELS EXPLAINED

- enforce : Sets the policy level (privileged|baseline|restricted)
- warn : Warns when a pod violates the level
- audit : Logs violations without blocking
- enforce-version : Policy version (use 'latest' for current)

ENFORCEMENT MODES

ENFORCE

Rejects pods that violate the policy.



Prevents non-compliant pods from running.

WARN

Allows pods but sends warnings.



Helps teams fix issues without breaking apps.

AUDIT

Allows pods and records violations.



Visibility and monitoring without enforcement.

MIGRATION FROM POD SECURITY POLICIES (PSP)

PSP (DEPRECATED)

- Complex and hard to manage.
- Cluster-wide resources.
- Many fields, difficult to maintain.
- Removed in Kubernetes 1.25+.



MIGRATION STEPS

- 1 Inventory existing PSPs and their rules.
- 2 Map rules to Pod Security Standards levels (Privileged, Baseline, Restricted).
- 3 Label namespaces with appropriate Pod Security Standard levels.
- 4 Test workloads and fix violations.
- 5 Remove PSPs and related RBAC.

PSA (POD SECURITY ADMISSION)

- Built-in, simple, and consistent.
- Namespace scoped via labels.
- Easier to understand and operate.
- Future of Kubernetes security.



BEST PRACTICES

DEFAULT TO RESTRICTED



Use Restricted for sensitive and multi-tenant workloads.

UNDERSTAND YOUR WORKLOADS



Know what your applications need. Avoid over-permissive policies.

USE WARN & AUDIT BEFORE ENFORCE



Start with warn/audit to find issues before enforcing policies.

KEEP SYSTEM NAMESPACES SAFE



Use appropriate levels for kube-system and platform namespaces.

CONTINUOUSLY MONITOR



Monitor audit logs and fix violations regularly.

AUTOMATE COMPLIANCE



Use GitOps, policy as code, and CI/CD checks to enforce standards.



COMMON VIOLATIONS

- Running as root
- Privileged containers
- HostPath volumes
- Allowing privilege escalation
- Adding dangerous capabilities



VERIQTA PRODUCTION NOTE

Pod Security Standards provide a strong, built-in security baseline for Kubernetes. Enforce the most restrictive level your workloads can tolerate.



KEY TAKEAWAY

Secure pods = Secure cluster. Use the right level, enforce it at the namespace level, and continuously improve.



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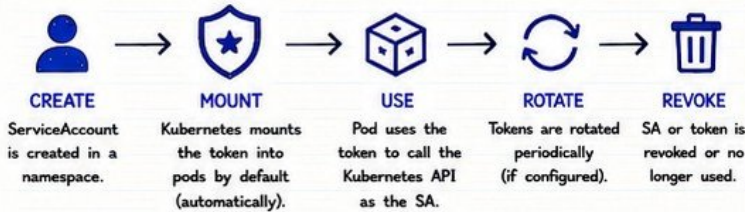
Service Accounts give workloads an identity. Tokens give access. Protect them like passwords. Rotate them often.

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8. SERVICE ACCOUNTS AND TOKEN SECURITY

IDENTITY FOR WORKLOADS. SECURE BY DESIGN.

1. SERVICE ACCOUNT LIFECYCLE



ServiceAccounts exist within a namespace and follow RBAC permissions. They are not automatically granted any rights.

2. PROJECTED TOKENS (RECOMMENDED)

Use projected service account tokens instead of legacy long-lived tokens.



- ✓ Short-lived
- ✓ Bound to audience
- ✓ Bound to pod (UID)
- ✓ Auto-rotated by Kubelet
- ✓ Stored in memory
- ✓ Not persisted on disk
- ✓ Safer than legacy secrets

```
# Example: projected token volume (default in 1.24+)
volumes:
- name: kube-api-access
  projected:
    sources:
    - serviceAccountToken:
        audience: https://kubernetes.default.svc
        expirationSeconds: 3600
        path: token
```

3. TOKEN ROTATION



- ✓ Kubelet automatically rotates tokens before they expire.
- ✓ Default token TTL: 1 hour (configurable).
- ✓ Tokens are short-lived and audience-bound.
- ✓ Old tokens are invalidated automatically.
- ✓ No manual intervention required.

4. TOKEN MISUSE AND RISKS

- ⚠ Long-lived tokens can be stolen and reused.
- ⚠ Tokens with excessive RBAC permissions increase blast radius.
- ⚠ Tokens mounted in hostPath or logged by applications can leak.
- ⚠ Tokens used outside the intended audience (kube-api) are dangerous.
- ⚠ ServiceAccount tokens in container images are a critical risk.



TREAT TOKENS LIKE PASSWORDS. PROTECT THEM LIKE SECRETS.

5. SECURE CONFIGURATION

- ✓ Disable automatic mounting of SA tokens.
- ✓ Use least privilege RBAC for each Service Account.
- ✓ Use projected tokens (BoundServiceAccountTokenVolume = true).
- ✓ Set short token TTL via TokenRequest API.
- ✓ Use audience restriction.
- ✓ Avoid using the default Service Account.
- ✓ Monitor and audit token usage.
- ✓ Rotate tokens regularly (automatic rotation recommended).
- ✓ Never bake tokens into container images.

HOW TO DISABLE AUTOMATIC TOKEN MOUNTING

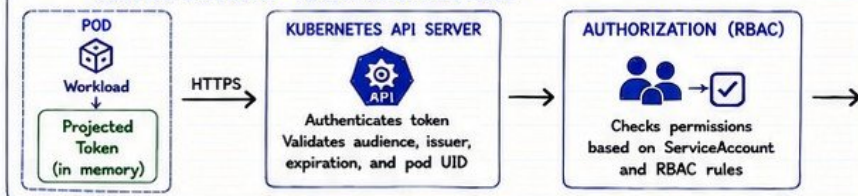
```
# At Namespace level
apiVersion: v1
kind: Namespace
metadata:
  name: secure-apps
  labels:
    pod-security.kubernetes.io/enforce: restricted

# At Pod level
spec:
  automountServiceAccountToken: false
```

NOTE

Pods only get a token if they need it. Disable by default.

SERVICE ACCOUNT TOKEN ARCHITECTURE



TOKEN TYPES COMPARISON

TYPE	STORAGE	TTL	BOUND	RECOMMENDED
Legacy Token	Secret (etc.)	Long-lived	No	✗
Projected Token	Memory	Short-lived (1h default)	Yes (pod, audience)	✓

COMMON MISTAKES

- ✗ Using default Service Account.
- ✗ Mounting tokens in every pod.
- ✗ Over-permissive RBAC roles.
- ✗ Storing tokens in images or code.
- ✗ Ignoring token rotation and audit.



VERIQTA PRODUCTION NOTE

Secure Service Account tokens reduce risk, limit blast radius, and protect the control plane. Use short-lived, audience-bound tokens and least-privilege access.



KEY TAKEAWAY

Identity for workloads. Tokens for access. Secure by design. Rotate and monitor always.



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9. SECRETS MANAGEMENT

SECURE YOUR SECRETS. PROTECT YOUR CLUSTER.



VERIQTA TIP

Secrets are sensitive.
Handle them with
care. Everyone with
access to a secret
can use it.

1. KUBERNETES SECRETS

Kubernetes Secrets store sensitive information such as passwords, tokens, TLS keys, and API keys.
Secrets are base64 encoded, **not encrypted by default**.



CREATE

kubectl create
secret generic
my-secret --from-literal
password=MyPass123



SECRET

Stored in etcd
(base64 encoded)
Not encrypted
by default



POD

Mounted as
environment variable
or volume

SECRET TYPES

- Opaque (default)
- [kubernetes.io/tls](https://kubernetes.io/docs/concepts/configuration/secret/#opaque-secret)
- [kubernetes.io/dockerconfigjson](https://kubernetes.io/docs/concepts/configuration/secret/#dockerconfigjson-secret)
- Service Account Token (auto-generated)

USE CASES

- ✓ App passwords
- ✓ DB credentials
- ✓ API tokens
- ✓ TLS certificates
- ✓ OAuth client secrets

2. SECRET ENCRYPTION

Enable encryption at rest to protect Secrets in etcd.

ENCRYPTION AT REST FLOW



Secret
(Plain Text)



Encryption
Provider



etcd
(Encrypted)

How to enable:

```
# In kube-apiserver manifest
--encryption-provider-config=/etc/kubernetes/encryption.yaml
```

ENCRYPTION PROVIDERS (EXAMPLE)

PROVIDER	TYPE	NOTES
aescbc	Encryption at rest	Simple, built-in
kms	KMS (Cloud KMS)	Recommended for production
secretbox	Encryption at rest	Deprecated (legacy)



Always use KMS (e.g., AWS KMS, GCP KMS, Azure Key Vault)
for production-grade clusters.

3. EXTERNAL SECRET OPERATORS

External Secret Operators sync secrets from external systems into Kubernetes.

OPERATOR	SOURCE SYSTEMS	FEATURES
External Secrets 	AWS Secrets Manager HashiCorp Vault Azure Key Vault GCP Secret Manager ...	• Sync to K8s Secret • Auto refresh • Scoped access • Templating
Sealed Secrets 	GitOps friendly (Helm / Kustomize / ArgoCD)	• Encrypt once • Safe in Git • Controller decrypts
Vault Secrets Operator 	HashiCorp Vault	• Dynamic secrets • Leasing • Rotation support

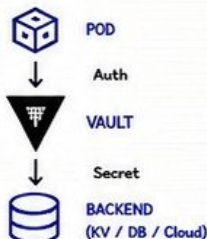


Best practice: Do not store real secrets in Git.
Use External Secrets or Sealed Secrets.

4. VAULT INTEGRATION

HashiCorp Vault provides centralized secret storage, dynamic secrets, and fine-grained access control.

ARCHITECTURE



INTEGRATION METHODS

- Vault Agent Injector
 - Injects secrets as env vars or files
 - Auto renew & rotate
- CSI Secrets Store Driver
 - Mount secrets as files
 - Sync with Vault
- Vault API
 - Apps fetch secrets directly
 - Use short-lived tokens



Use Vault policies to enforce least privilege.
Never give broad access to applications.

5. SECRET ROTATION

Rotate secrets regularly to reduce risk of exposure.

- ✓ Use short-lived credentials
- ✓ Enable automatic rotation (e.g., DB, API keys)
- ✓ Vault dynamic secrets where possible
- ✓ Rotate Service Account tokens
- ✓ Monitor and alert on expiring secrets



AUTOMATE IT.
DON'T FORGET IT.

6. COMMON MISTAKES

- ✗ Storing secrets in ConfigMaps
- ✗ Hardcoding secrets in code or images
- ✗ Committing secrets to Git
- ✗ Using default ServiceAccount tokens
- ✗ Not enabling encryption at rest
- ✗ Granting overly broad RBAC access to secrets
- ✗ Not rotating secrets regularly



ONE LEAKED SECRET
CAN COMPROMISE
YOUR ENTIRE CLUSTER.

BEST PRACTICES SUMMARY



Enable encryption
at rest (KMS).



Use least privilege
RBAC for secrets.



Rotate secrets
regularly.



Use external secret
managers (Vault,
AWS SM, etc.).



Never store real
secrets in Git.



Audit access and
monitor usage.



VERIQTA PRODUCTION NOTE

Secrets are the keys to your kingdom.
Protect them like your business depends on it—because it does.



KEY TAKEAWAY

Secure storage, controlled access,
and automated rotation keep your
workloads and data safe.



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10. ENCRYPTING CLUSTER DATA

ENCRYPT EVERYTHING THAT MATTERS.

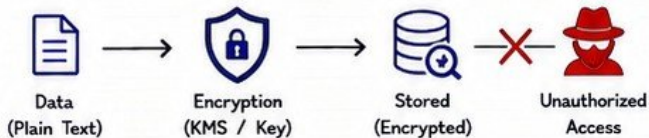


VERIQTA TIP

If it's not encrypted,
assume it's exposed.
Encrypt at rest.
Protect in transit.
Secure by design.

1. ENCRYPTION AT REST OVERVIEW

Encryption at rest protects data stored on disk from unauthorized access.



Protect:

- ✓ etcd data
- ✓ Secrets
- ✓ ConfigMaps (sensitive)
- ✓ Persistent Volumes
- ✓ Backups
- ✓ Logs (if sensitive)

3. KMS PROVIDERS

Use a Key Management Service (KMS) to manage encryption keys.

PROVIDER	TYPE	NOTES
AWS KMS	Cloud KMS	Highly available, integrated with IAM
Azure Key Vault	Cloud KMS	Managed keys, RBAC, audit logs
Google Cloud KMS	Cloud KMS	Regional keys, IAM integration
HashiCorp Vault	Self-managed	Dynamic secrets, fine-grained control
KMIP	External KMS	Standard protocol for HSM/KMS

- ✓ Use least-privilege IAM/RBAC for KMS access.

5. BACKUP PROTECTION

Backups contain sensitive data. Encrypt them.

BACKUP FLOW



PROTECT BACKUPS

- ✓ Encrypt backups before storing
- ✓ Store in secure, versioned locations (S3, GCS, etc.)
- ✓ Restrict bucket/container access
- ✓ Enable bucket encryption
- ✓ Test restore regularly

- ✓ Backups without encryption can expose your entire cluster.

7. COMMON MISTAKES

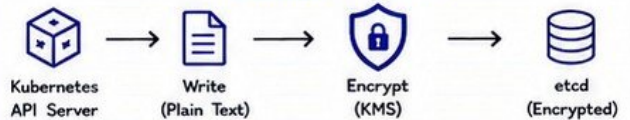
- ✗ Not enabling etcd encryption
- ✗ Using identity provider (no encryption)
- ✗ Storing keys in ConfigMaps or files
- ✗ Not rotating keys
- ✗ Overly broad KMS permissions
- ✗ Unencrypted backups
- ✗ Losing access to KMS keys
- ✗ Not testing backup restore



2. ETCD ENCRYPTION

etcd stores the entire cluster state. Encrypt it.

ETCD ENCRYPTION FLOW



Enable etcd encryption provider in the API server.

```
# Example: /etc/kubernetes/manifests/kube-apiserver.yaml
...
--encryption-provider-config=/etc/kubernetes/encryption.yaml
...
```

ENCRYPTION PROVIDER CONFIG EXAMPLE

```
apiVersion: apiserver.config.k8s.io/v1
kind: EncryptionConfiguration
resources:
- resources: [secrets]
  providers:
  - kms:
    name: aws-kms
    endpoint: unix:///var/run/kmsplugin/socket.sock
    identity: {} # fallback (no encryption)
```

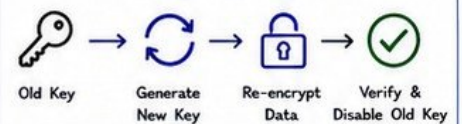
4. KEY ROTATION

Rotate keys regularly to reduce risk if a key is compromised.

ROTATION STRATEGY

- ✓ Rotate KMS keys (e.g., every 90 days)
- ✓ Use automatic rotation where available
- ✓ Re-encrypt data with new key
- ✓ Test decryption after rotation
- ✓ Keep old keys enabled until re-encryption is complete

ROTATION WORKFLOW



Never delete old keys immediately.
Keep them for decryption until migration is complete.

6. BEST PRACTICES



Encrypt etcd at rest



Use KMS, not static keys



Rotate keys regularly



Use least privilege for KMS access



Audit encryption and key usage



Protect backups and offsite copies



VERIQTA PRODUCTION NOTE

Encryption is not optional.
It's your last line of defense
when everything else fails.



KEY TAKEAWAY

Encrypt etcd, use KMS,
rotate keys, and protect
your backups.
Secure the data. Secure
the cluster.



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11. API SERVER SECURITY

SECURE THE CONTROL PLANE. SECURE THE CLUSTER.

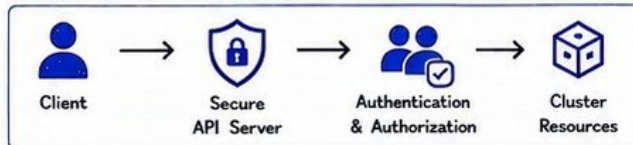


VERIQTA TIP

The API server is the front door to your cluster. Lock it down.

1. SECURE CONFIGURATION OVERVIEW

The API server exposes the Kubernetes control plane. Secure configuration prevents unauthorized access, privilege escalation, and data leaks.



GOALS:

- ✓ Authenticate every request
- ✓ Authorize least privilege access
- ✓ Encrypt all traffic
- ✓ Audit every important action
- ✓ Disable unnecessary access
- ✓ Harden the API server

3. AUDIT LOGGING

Audit logs record API requests for security, compliance, and troubleshooting.

AUDIT LEVELS

LEVEL	DESCRIPTION	USE CASE
None	No audit logging	Not recommended
Metadata	Log metadata of requests	Minimal logging
Request	Log requests (no bodies)	General auditing
RequestResponse	Log requests and responses	Security & forensic

BEST PRACTICES

- ✓ Use Request or RequestResponse in production.
- ✓ Store logs securely and centrally.
- ✓ Protect audit logs from tampering and deletion.
- ✓ Monitor for suspicious activities.

5. SECURE FLAGS (IMPORTANT)

Use secure flags to harden the API server.

```
--anonymous-auth=false
--insecure-port=0
--authorization-mode=Node,RBAC
--enable-admission-plugins=NodeRestriction,...
--audit-log-path=/var/log/kubernetes/audit.log
--audit-policy-file=/etc/kubernetes/audit-policy.yaml
--tls-cert-file=/etc/kubernetes/pki/apiserver.crt
--tls-private-key-file=/etc/kubernetes/pki/apiserver.key
--client-ca-file=/etc/kubernetes/pki/ca.crt
--etcd-servers=https://127.0.0.1:2379
--etcd-cafile=/etc/kubernetes/pki/etcd/ca.crt
--etcd-certfile=/etc/kubernetes/pki/apiserver-etcd-client.crt
--etcd-keyfile=/etc/kubernetes/pki/apiserver-etcd-client.key
```

CRITICAL FLAGS

Never expose the API server on public networks. Use firewalls and security groups.

COMMON MISTAKES

- ✗ Leaving anonymous access enabled
- ✗ Using insecure port or HTTP
- ✗ Not enabling audit logging
- ✗ Overly permissive RBAC policies
- ✗ Exposing API server to the internet
- ✗ Not rotating certificates



VERIQTA PRODUCTION NOTE

The API server is the most critical component in Kubernetes. If it is compromised, the entire cluster is compromised. Harden it like a fortress.



KEY TAKEAWAY

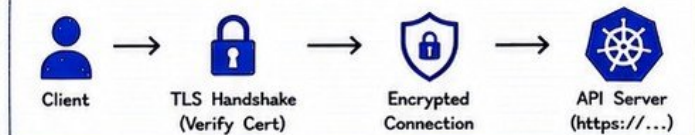


Secure the API server with strong authentication, TLS, audit logs, and strict access controls. Harden today. Protect forever.

2. TLS (TRANSPORT LAYER SECURITY)

Encrypt all communication to the API server using TLS.

TLS FLOW



BEST PRACTICES

- ✓ Use strong TLS versions (1.2 or 1.3)
- ✓ Use trusted CA-signed certificates
- ✓ Use SANs for all API server endpoints
- ✓ Enable client certificate authentication
- ✓ Disable insecure cipher suites

CHECK YOUR CERT

```
$ openssl s_client -connect
<apiserver>:6443
$ openssl x509 -in server.crt
-noout -text
```



4. ANONYMOUS ACCESS

Disable anonymous access unless absolutely required.

ANONYMOUS ACCESS RISKS

- Information disclosure
- Enumeration of resources
- Attack surface expansion
- Privilege escalation paths



BEST PRACTICES

- ✓ Set --anonymous-auth=false
- ✓ Do not bind to 0.0.0.0 without firewalls
- ✓ Restrict access to trusted networks only
- ✓ Use RBAC for all authenticated users

EXAMPLE

```
# Disable anonymous access
--anonymous-auth=false

# Disable insecure port
--insecure-port=0
```

6. API HARDENING CHECKLIST

AREA	ACTION	STATUS
Authentication	Enable strong auth (certs, tokens, OIDC)	☐
Authorization	Use RBAC with least privilege	☐
TLS	Use TLS 1.2+ and valid certificates	☐
Audit	Enable audit logging (Request or higher)	☐
Anonymous Access	Disable anonymous access	☐
Network	Restrict API server network access	☐
Admission	Enable security admission plugins	☐
Secrets & Data	Encrypt etcd and protect backups	☐



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12. ETCD SECURITY

SECURE THE HEART OF YOUR CLUSTER.



VERIQTA TIP

etcd stores the most critical cluster state. Protect it like your most valuable asset. Always secure, always back up, always test.

1. TLS COMMUNICATION (ENCRYPT IN TRANSIT)

etcd uses TLS to encrypt all communication between members and clients.



BEST PRACTICES

- ✓ Use TLS for client and peer communication
- ✓ Use strong TLS versions (TLS1.2 or TLS1.3)
- ✓ Use certificates issued by trusted CA
- ✓ Rotate certificates before they expire

4. ENCRYPTION

Encrypt data at rest to protect sensitive cluster information.



NOTES

Use etcd encryption at rest feature with a KMS provider. Keys should never be stored on etcd nodes.

2. AUTHENTICATION

Enable authentication to control who can access etcd.

METHODS

- Client Authentication with certificates
- etcdctl username/password authentication (basic auth)
- Role-Based Access Control (RBAC) with etcd v3

3. BACKUP SECURITY

Backups must be protected and verifiable.

- ✓ Encrypt all backups
- ✓ Store backups in secure, access-controlled locations
- ✓ Verify backup integrity regularly
- ✓ Test restore procedures

5. DISASTER RECOVERY

Plan and test your etcd disaster recovery strategy.

- ✓ Regular backups and snapshots
- ✓ Multi-member cluster across zones or regions
- ✓ Tested restore and failover procedures
- ✓ Document recovery steps clearly

6. OPERATIONAL BEST PRACTICES



Monitor etcd health

Watch metrics and alerts



Keep cluster healthy

Maintain majority quorum



Regular upgrades

Keep etcd and components up to date



Limit access

Use least privilege for etcd access



Protect keys

Secure KMS and TLS certificates



Audit and review

Review logs and access regularly

ETCD SECURITY CONFIG EXAMPLE

```
# etcd cluster configuration
etcd:
  name: etcd-node-1
  configured-peers: ["https://10.0.0.1:2380","https://10.0.0.2:2380","https://10.0.0.3:2380"]
  listen-client-urls: https://0.0.0.0:2379
  advertise-client-urls: https://10.0.0.1:2379
  listen-peer-urls: https://0.0.0.0:2380
  initial-cluster: "etcd-node-1=https://10.0.0.1:2380,etcd-node-2=https://10.0.0.2:2380,etcd-node-3=https://10.0.0.3:2380"
  auto-tls: true
  cert-file: /etc/kubernetes/pki/etcd/server.crt
  key-file: /etc/kubernetes/pki/etcd/server.key
  client-cert-auth: true
  peer-client-cert-auth: true
  encryption: enabled
  encryption-provider: aescbc
  data-dir: /var/lib/etcd
  snapshot-count: 10000
```

KEY TAKEAWAY



etcd security protects the control plane. Small missteps can cause big outages.

COMMON MISTAKES

- ✗ Running etcd without TLS
- ✗ Storing unencrypted backups
- ✗ Weak or unmanaged certificates
- ✗ Open etcd client endpoints
- ✗ No monitoring or backups
- ✗ Ignoring disaster recovery testing



SECURE etcd. PROTECT YOUR CLUSTER. SAFE TODAY, STRONG TOMORROW.



PROTECT
BACKUP
RECOVER



A SECURE etcd IS A
RESILIENT CLUSTER



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VERIQTA TIP

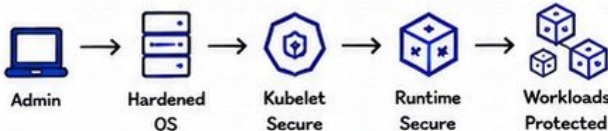
If a node is compromised, the attacker can access your entire cluster. Harden every layer.

13. NODE SECURITY

HARDEN YOUR NODES. PROTECT YOUR WORKLOADS.

1. WORKER NODE HARDENING OVERVIEW

Worker nodes run your workloads. Harden every layer from the OS to the container runtime.



GOALS:

- ✓ Reduce attack surface
- ✓ Prevent privilege escalation
- ✓ Protect node credentials
- ✓ Isolate workloads
- ✓ Maintain availability
- ✓ Enable detection & response

2. OPERATING SYSTEM SECURITY

Secure and harden the underlying operating system.

AREA	BEST PRACTICES
Minimal OS	Use minimal, supported OS (e.g., Ubuntu LTS, Amazon Linux 2)
Disable Unused Services	Disable unnecessary services and daemons
Firewall	Allow only required ports (kubelet, container runtime, CNI)
SELinux / AppArmor	Enforce SELinux (preferred) or AppArmor profiles
File Permissions	Restrict access to system files and sensitive directories
User Accounts	Use non-root users for admin tasks
Sysctl Hardening	Apply secure sysctl settings (kernel hardening)
Logging	Forward logs to centralized logging system

EXAMPLE SECURE SYSCTL SETTINGS

```
net.ipv4.ip_forward = 0
net.ipv4.conf.all.rp_filter = 1
kernel.kptr_restrict = 2
fs.protected_hardlinks = 1
fs.protected_symlinks = 1
```

3. SECURE KUBELET

Configure kubelet securely to prevent unauthorized access and abuse.

KUBELET
BEST
PRACTICES

- ✓ Use TLS authentication for kubelet API
- ✓ Disable anonymous authentication
- ✓ Use strong authorization mode (Webhook)
- ✓ Rotate kubelet client certificates regularly
- ✓ Restrict read-only port (or disable it)
- ✓ Protect kubelet config and certificates
- ✓ Enable node authorization
- ✓ Use systemd drop-ins for secure flags

SECURE KUBELET FLAGS (EXAMPLE)

```
--anonymous-auth=false
--client-ca-file=/etc/kubernetes/pki/ca.crt
--tls-cert-file=/etc/kubernetes/pki/kubelet.crt
--tls-private-key-file=/etc/kubernetes/pki/kubelet.key
--authorization-mode=Webhook
--read-only-port=0
--protect-kernel-defaults=true
```

4. SSH RESTRICTIONS

Limit SSH access. Use it only when absolutely necessary.

DO THIS

- ✓ Use key-based authentication
- ✓ Allow SSH from specific IPs only
- ✓ Use a bastion host / jump server
- ✓ Use non-root user
- ✓ Disable password authentication
- ✓ Set client alive and timeouts

DON'T DO THIS

- ✗ Don't allow SSH from anywhere
- ✗ Don't use root login
- ✗ Don't use shared SSH accounts
- ✗ Don't leave default SSH configs
- ✗ Don't expose SSH on public IP
- ✗ Don't forget to log and monitor

SSH HARDENING (/etc/ssh/sshd_config)

```
PermitRootLogin no
PasswordAuthentication no
ChallengeResponseAuthentication no
AllowUsers admin
AllowGroups sshusers
MaxAuthTries 3
ClientAliveInterval 300
ClientAliveCountMax 2
```



5. PATCH MANAGEMENT

Keep your nodes and software up to date.

PATCH STRATEGY

- ✓ Enable automatic security updates for OS packages
- ✓ Patch container runtime, kubelet, kubectl, and CNI plugins
- ✓ Schedule regular maintenance windows
- ✓ Test updates in staging environment
- ✓ Monitor vendor security advisories

TOOLS YOU CAN USE

- ✓ unattended-upgrades (Linux)
- ✓ yum / dnf / apt
- ✓ AWS SSM Patch Manager
- ✓ Ansible / Puppet / Chef
- ✓ Kubernetes Node Problem Detector



Unpatched nodes are the easiest entry point for attackers. Patch early. Patch often.

COMMON MISTAKES

- ✗ Running nodes with default or weak configs
- ✗ Leaving kubelet read-only port enabled
- ✗ Using root SSH access
- ✗ Not patching nodes regularly
- ✗ Exposing unnecessary ports
- ✗ Ignoring runtime security

6. RUNTIME ISOLATION

Isolate workloads and reduce impact of node compromise.

CONTAINER RUNTIME SECURITY

- Use containerd or CRI-O (avoid deprecated runtimes)
- Enable runtime security features (seccomp, AppArmor, SELinux)
- Disable privileged mode by default
- Block unnecessary Linux capabilities
- Use read-only root filesystem where possible
- Enable image signature verification (e.g., cosign, Notary v2)



ISOLATION TECHNIQUES



Run as non-root



Use SecurityContext



Enable Seccomp



Enable AppArmor/SELinux



Network Policies



VERIQTA PRODUCTION NOTE

A secure node is a strong node. Secure the OS, kubelet, runtime, and access. One weak node can compromise the entire cluster.



KEY TAKEAWAY

Harden every layer. Limit access. Keep systems updated. Isolate workloads. Monitor everything.



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VERIQTA TIP

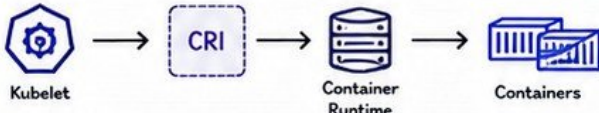
The container runtime is the engine that runs your containers. Secure the engine, secure everything.

14. CONTAINER RUNTIME SECURITY

SECURE YOUR RUNTIME. PROTECT YOUR CONTAINERS.

1. CONTAINER RUNTIME OVERVIEW

Kubernetes uses a CRI (Container Runtime Interface) to communicate with container runtimes.



Popular Runtimes:



High performance, mature, widely used.



CRI-O

Lightweight, secure, Kubernetes-native.

3. RUNTIME ISOLATION

Isolation prevents containers from breaking out and accessing the host or other containers.



Namespaces
(Process, Network,
Mount, IPC, UTS)



Cgroups
(CPU, Memory,
IO limits)



Read-only Root FS
Limit write
capabilities



User Namespaces
Map container
users safely



KEY GOAL:

Each container runs in a limited, isolated environment.

5. APPARMOR

AppArmor restricts what resources a container can access on the host.

EXAMPLE: AppArmor Profile

```
profile k8s-container flags=(attach_disconnected) {  
  capability,  
  file,  
  network,  
  deny /** *,  
  /var/log/** r,  
}
```



APPARMOR



BEST PRACTICE: Use the runtime/default profile or a custom AppArmor profile for additional restrictions.

7. RUNTIME HARDENING

Harden your runtime configuration to reduce risks.

CONTAINERD HARDENING

- ✓ Disable insecure registries
- ✓ Use TLS for registry access
- ✓ Set "no_new_privileges = true"
- ✓ Drop all unnecessary capabilities
- ✓ Use read-only root filesystem
- ✓ Limit PIDs, CPU & Memory



CRI-O HARDENING

- ✓ Use read-only root filesystem
- ✓ Enable seccomp by default
- ✓ Drop all capabilities
- ✓ Disable host PID/IPC
- ✓ Set user namespace remap
- ✓ Use minimal base images

COMMON MISTAKES

- ✗ Running privileged containers
- ✗ Disabling seccomp/AppArmor/SELinux
- ✗ Using insecure registries
- ✗ Allowing host PID/IPC
- ✗ Not limiting resources
- ✗ Using outdated runtime versions



VERIQTA PRODUCTION NOTE

The container runtime is the last line of defense before the host. Harden it well. One weak setting can break the cluster.



KEY TAKEAWAY

A secure runtime ensures containers stay isolated, systems stay protected, and attackers stay out.

2. RUNTIME COMPARISON

FEATURE	containerd	CRI-O
Type	General purpose runtime	Kubernetes-native runtime
Performance	High	High (lightweight)
Security	Strong	Strong (minimal attack surface)
Image Management	Built-in	Built-in (via containers/image)
Pod Sandbox	runc / runsc / gVisor	runc / crun
Default in Kubernetes	Yes (most distros)	Yes (in some distros)

BEST PRACTICES

- ✓ Use the latest stable version.
- ✓ Enable runtime security features.
- ✓ Use minimal images.
- ✓ Monitor runtime activity.

4. SECCOMP (SECURE COMPUTING MODE)

Seccomp reduces the attack surface by limiting system calls.

EXAMPLE: Default Seccomp Profile

Allows safe syscalls:
✓ read, write, exit, futex

Blocks dangerous syscalls:
✗ ptrace, kexec_load, keyctl

ENABLE SECCOMP (Pod Security Context)

```
securityContext:  
  seccompProfile:  
    type: RuntimeDefault  
# or use a custom profile  
# localhost/profile.json
```



BEST PRACTICE: Use RuntimeDefault or a custom seccomp profile for all workloads.

6. SELINUX

SELinux provides mandatory access control (MAC) for containers.

MODES

- Enforcing - Fully enforced (recommended)
- Permissive - Logs violations, allows actions
- Disabled - No protection (not recommended)

CHECK STATUS

```
sestatus  
getenforce
```

ENABLE SELINUX FOR CONTAINERS

Match your runtime with the correct SELinux type.

- ✗ container_t - General container
- ✓ container_runtime_t - Runtime processes
- ✗ container_file_t - Container files



SELINUX

8. RUNTIME SECURITY CHECKLIST

- ✓ Use latest runtime version
- ✓ Enable seccomp (RuntimeDefault)
- ✓ Enable AppArmor profile
- ✓ Enable SELinux (Enforcing)
- ✓ Drop all unnecessary capabilities
- ✓ Use read-only root filesystem
- ✓ Disable privileged containers
- ✓ Restrict host PID/IPC access
- ✓ Limit resources (CPU/Memory/PID)
- ✓ Use minimal base images
- ✓ Enable user namespaces
- ✓ Monitor runtime events and logs

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15. POD SECURITY CONTEXT

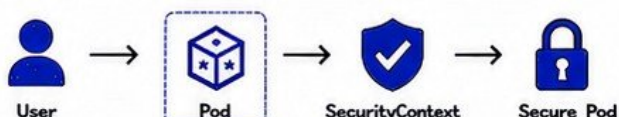
LOCK DOWN YOUR PODS. LIMIT PRIVILEGES. PREVENT ESCALATION.

VERIQTA TIP

The principle of least privilege starts at the pod level. Configure SecurityContext for every container.

1. WHAT IS POD SECURITY CONTEXT?

SecurityContext defines privilege and access settings for pods and containers. It helps enforce least privilege and reduces attack impact.



KEY BENEFITS

- ✓ Prevents privilege escalation
- ✓ Reduces container breakout risk
- ✓ Enforces least privilege
- ✓ Limits damage from exploits
- ✓ Improves workload isolation
- ✓ Meets compliance requirements

3. RUN AS NON-ROOT

Running containers as non-root prevents attackers from getting root access inside the container.

DO THIS

- ✓ Run as specific non-root user
- ✓ Set runAsNonRoot: true
- ✓ Define runAsUser

DON'T DO THIS

- ✗ Running as root (UID 0)
- ✗ Allowing root group access
- ✗ Leaving defaults undefined

EXAMPLE (Pod SecurityContext)

```
securityContext:
  runAsNonRoot: true
  runAsUser: 1000
  runAsGroup: 10001
  fsGroup: 10001
```

5. CAPABILITIES MANAGEMENT

Linux capabilities give specific privileges. Drop all unnecessary capabilities to reduce risk.

DO THIS

- ✓ Drop ALL capabilities and add only what is needed
- ✓ Use drop: ["ALL"]
- ✓ Add specific caps if required

DON'T DO THIS

- ✗ Allow privileged containers
- ✗ Keep default capabilities
- ✗ Add unnecessary capabilities

EXAMPLE (Drop All & Add Specific)

```
securityContext:
  capabilities:
    drop:
      - ALL
    add:
      - NET_BIND_SERVICE
      - CHOWN
```

COMMON CAPABILITIES TO AVOID

- ✗ SYS_ADMIN - Full system access
- ✗ NET_ADMIN - Network admin access
- ✗ SYS_MODULE - Load kernel modules
- ✗ DAC_OVERRIDE - Bypass file permissions
- ✗ SETUID - Change user ID

COMMON MISTAKES

- ✗ Running containers as root
- ✗ Not setting SecurityContext
- ✗ Allowing privilege escalation
- ✗ Not dropping capabilities
- ✗ Writing to root filesystem



VERIQTA PRODUCTION NOTE

Attackers target weak pods first. A strong SecurityContext blocks privilege escalation and limits the blast radius.



KEY TAKEAWAY

Secure every pod by enforcing least privilege. Use non-root users, drop capabilities, disable escalation, and keep filesystems read-only.

2. SECURITYCONTEXT OVERVIEW

SecurityContext can be set at the Pod level (applies to all containers) or at the Container level (overrides Pod level).

POD LEVEL

Applies to all containers in the pod.



CONTAINER LEVEL

Overrides pod settings for a specific container.



BEST PRACTICES

- ✓ Set SecurityContext for every Pod and Container.
- ✓ Run as non-root user.
- ✓ Use read-only root filesystem whenever possible.
- ✓ Drop all unnecessary capabilities.
- ✓ Disable privilege escalation.
- ✓ Test and validate security settings.

4. READ-ONLY ROOT FILESYSTEM

Make the container filesystem read-only to prevent changes and persistence of malicious activity.

DO THIS

- ✓ Set readOnlyRootFilesystem: true
- ✓ Mount writable volumes only where needed
- ✓ Use tmpfs for temporary data

DON'T DO THIS

- ✗ Allow write access to root filesystem
- ✗ Store data in container layer
- ✗ Use unnecessary writable paths

EXAMPLE (Container SecurityContext)

```
securityContext:
  readOnlyRootFilesystem: true
  volumes:
    - name: tmp
      emptyDir:
        medium: Memory
  volumeMounts:
    - name: tmp
      mountPath: /tmp
```

6. PRIVILEGE ESCALATION & USER NAMESPACES

DISABLE PRIVILEGE ESCALATION

Prevents processes from gaining more privileges (e.g., via setuid binaries).

```
securityContext:
  allowPrivilegeEscalation: false
```



USER NAMESPACES (KUBERNETES 1.25+)

Maps container root to non-root on the host. Adds an extra layer of isolation.

```
securityContext:
  userNamespaceMode: "Auto"
```



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16. LINUX KERNEL SECURITY FEATURES

THE KERNEL IS YOUR LAST LINE OF DEFENSE.
SECURE THE FOUNDATION.

VERIQTA TIP

The Linux kernel is the foundation of your security. Harden it, isolate everything, and reduce the attack surface.

1. LINUX KERNEL SECURITY OVERVIEW

These kernel features work together to isolate workloads, limit capabilities, and reduce the potential attack surface.



KEY BENEFITS

- ✓ Strong isolation between containers and processes
- ✓ Least privilege and restricted capabilities
- ✓ Minimized kernel attack surface
- ✓ Better containment of compromised workloads
- ✓ Foundation for defense in depth

3. CGROUPS (RESOURCE CONTROL)

Cgroups limit, account and isolate resource usage for containers.



- ✓ CPU shares and limits
- ✓ Memory limits and OOM control
- ✓ Block IO throttling
- ✓ Network bandwidth control
- ✓ PIDs limit
- ✓ Device access control

RESULT: Prevents noisy neighbor problems and resource exhaustion attacks.

5. APPARMOR (MAC - LSM)

AppArmor is a Linux Security Module that enforces mandatory access control using profiles.



- ✓ Path-based access control
- ✓ Restrict file read/write execution
- ✓ Network and capability restrictions
- ✓ Enforce least privilege
- ✓ Per-container profile enforcement

EXAMPLE APPARMOR PROFILE RULE

```
/usr/bin/nginx ix,  
/var/www/** x,  
/var/log/nginx/** rw,  
network inet stream,  
deny /** w,
```

7. KERNEL ATTACK SURFACE MANAGEMENT

Reduce the kernel attack surface to limit potential exploits.



- ✓ Keep kernel up to date (security patches)
- ✓ Disable unused kernel modules
- ✓ Remove unnecessary features (CONFIG_*)
- ✓ Use minimal capabilities and seccomp
- ✓ Restrict /proc, /sys and debug interfaces
- ✓ Monitor kernel logs and anomalies

RESULT: A smaller attack surface means fewer opportunities for attackers.

COMMON MISTAKES

- ✗ Running containers as root
- ✗ Disabling AppArmor/SELinux
- ✗ Allowing privileged containers
- ✗ Not limiting capabilities



VERIQTA PRODUCTION NOTE

A secure kernel is the foundation of a secure platform. Harden the kernel, isolate everything, monitor continuously, and assume attackers are already inside.

- ✗ hritngn onocbrutis tiés
- ✗ Ignoring kernel updates and logs

2. NAMESPACES (ISOLATION)

Namespaces provide isolation of system resources so containers cannot see or affect each other.

NAMESPACE	ISOLATES
PID	Process ID space
NET	Network devices, stacks, ports
MNT	Filesystem mount points
IPC	System V IPC, POSIX queues, signals
UTS	Hostname and NIS domain name
USER	User and group ID space
CGROUP	Cgroup root directory

RESULT: Each container gets its own view of system resources. Breakout attempts are isolated by design.

4. SECCOMP (SECURE COMPUTING MODE)

Seccomp filters restrict the system calls a container can make. Only allowed syscalls are permitted.



ALLOW

- ✓ read
- ✓ write
- ✓ exit
- ✓ futex
- ✓ clock_gettime
- ✓ (etc)

DENY

- ✗ ptrace
- ✗ mount
- ✗ kexec_load
- ✗ keyctl
- ✗ (dangerous syscalls)

RESULT: Blocks entire classes of kernel exploits by reducing syscall attack surface.

6. SELINUX (MAC - LSM)

SELinux enforces mandatory access control at a fine-grained level using policies and labels.



- ✓ Type Enforcement (TE)
- ✓ Role-Based Access Control (RBAC)
- ✓ Least privilege access
- ✓ Process and file labeling
- ✓ Example modes: Enforcing, Permissive, Disabled

EXAMPLE SELINUX CONTEXT

system_u:system_r:container_t:s0:c123,c456

User Role Type Level Categories

KEY TAKEAWAY



Namespaces isolate. Cgroups control resources. Seccomp filters syscalls. AppArmor and SELinux enforce access control. Together, these Linux kernel features create a strong security shield for your containers and workloads.



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VERIQTA TIP

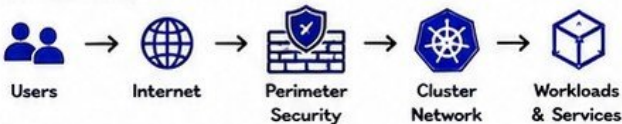
Your network is the highway of your cluster. Secure the path, control the traffic, protect the destination.

17. NETWORK SECURITY

SECURE YOUR NETWORK. PROTECT YOUR CLUSTER.

1. NETWORK SECURITY OVERVIEW

Kubernetes networks connect users, services, and workloads. Secure the architecture, control the traffic, and protect communication.



KEY GOALS

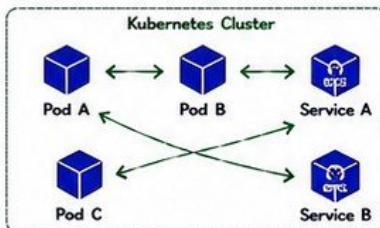
- ✓ Control traffic flow
- ✓ Encrypt communication
- ✓ Restrict access
- ✓ Reduce attack surface
- ✓ Protect workloads
- ✓ Prevent lateral movement
- ✓ Ensure secure exposure
- ✓ Monitor and detect threats

3. EAST-WEST TRAFFIC (INSIDE CLUSTER)

Traffic between workloads and services inside the cluster.

EXAMPLES

- Pod → Pod
- Pod → Service
- Service → Service
- Node → Pod



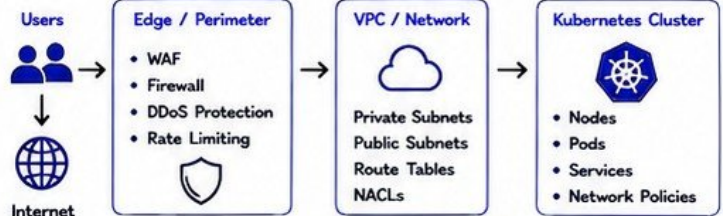
SECURITY CONTROLS

- ✓ Network Policies (allow only required traffic)
- ✓ Micro-segmentation
- ✓ Limit open ports between workloads
- ✓ Monitor internal traffic



2. NETWORK ARCHITECTURE

Secure network design is the foundation of cluster security.



SECURITY CONTROLS AT EVERY LAYER

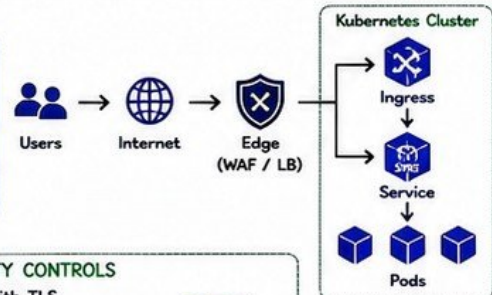


4. NORTH-SOUTH TRAFFIC (INTO / OUT OF CLUSTER)

Traffic between the cluster and external clients or services.

EXAMPLES

- User → Ingress
- Ingress → Service
- Pod → Internet
- Cluster → External API



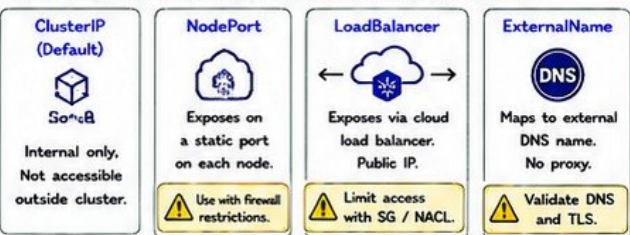
SECURITY CONTROLS

- ✓ Ingress controllers with TLS
- ✓ WAF and DDoS protection
- ✓ Restrict source IPs
- ✓ Egress filtering for outbound traffic



5. SERVICE EXPOSURE

Expose only what is necessary. Avoid unnecessary public access.



BEST PRACTICE: Prefer Ingress + Service (ClusterIP) for applications. Expose the minimum necessary.

COMMON MISTAKES

- ✗ Exposing services with NodePort / LoadBalancer unnecessarily
- ✗ No Network Policies (allow-all by default)
- ✗ No encryption for in-cluster communication
- ✗ Open egress to the internet
- ✗ Weak or missing TLS configuration



VERIQTA PRODUCTION NOTE

Attackers move laterally inside your cluster. Network segmentation and encryption are your best defense against lateral movement and data theft.

6. SECURE COMMUNICATION

Encrypt your traffic. Verify identity. Prevent eavesdropping and tampering.

IN-CLUSTER COMMUNICATION

- ✓ Enable TLS for all services
- ✓ Use mTLS between services
- ✓ Verify certificates
- ✓ Encrypt etcd and kube-apiserver communication

EXTERNAL COMMUNICATION

- ✓ Use HTTPS everywhere
- ✓ Enforce TLS 1.2+
- ✓ Use valid certificates
- ✓ HSTS for web applications

TOOLS & TECHNOLOGIES



KEY TAKEAWAY



Design a secure network architecture. Control east-west and north-south traffic. Expose only what is needed. Encrypt everything. Monitor always.



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VERIQTA TIP

If it is not explicitly allowed by a policy, it should not be able to communicate.

18. NETWORK POLICIES

CONTROL POD-TO-POD COMMUNICATION. ENFORCE ZERO TRUST.

1. WHAT ARE NETWORK POLICIES?

Network Policies control traffic flow between Pods and other network endpoints in Kubernetes.



KEY BENEFITS

- ✓ Restrict unwanted traffic
- ✓ Reduce attack surface
- ✓ Enable multi-tenant isolation
- ✓ Enforce least privilege
- ✓ Improve workload security
- ✓ Meet compliance requirements

2. INGRESS RULES (INCOMING TRAFFIC)

Controls which sources can connect to your Pods.

EXAMPLE: Allow traffic to your app only from the web namespace

```

apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-web-ingress
spec:
  podSelector:
    matchLabels:
      app: backend
  policyTypes:
    - Ingress
  ingress:
    - from:
        - namespaceSelector:
            matchLabels:
              name: web
  
```

KEY POINTS

- ✓ Defines allowed sources
- ✓ Based on Pods, namespaces or IP blocks
- ✓ Applied to selected Pods
- ✓ Multiple rules can be defined



3. EGRESS RULES (OUTGOING TRAFFIC)

Controls where your Pods can send traffic to.

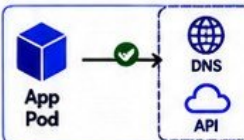
EXAMPLE: Allow DNS and API access only

```

apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-egress
spec:
  podSelector:
    matchLabels:
      app: backend
  policyTypes:
    - Egress
  egress:
    - to:
        - namespaceSelector:
            matchLabels:
              name: kube-system
      ports:
        - protocol: UDP
          port: 53
  
```

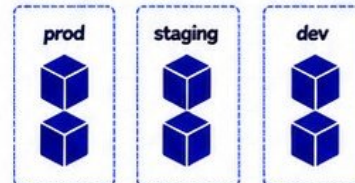
KEY POINTS

- ✓ Restricts outbound traffic
- ✓ Prevents data exfiltration
- ✓ Limits access to only required services
- ✓ Improves cluster security



4. NAMESPACE ISOLATION

Separate workloads by namespace and control communication.



STRATEGIES

- ✓ Use NamespaceSelectors
- ✓ Prevent cross-namespace access by default
- ✓ Allow only specific namespace communication
- ✓ Improve multi-tenant security

RESULT: Each namespace is isolated, and workloads communicate only when explicitly allowed.



5. DEFAULT DENY (ZERO TRUST)

Start with deny-all, then allow only what is needed.

DEFAULT DENY POLICY

```

apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny
spec:
  podSelector: {}
  policyTypes:
    - Ingress
    - Egress
  
```

WHY DEFAULT DENY?

- ✓ Blocks all traffic by default
- ✓ Eliminates unknown access
- ✓ Enforces explicit rules
- ✓ Prevents lateral movement
- ✓ Foundation for zero trust

COMMON MISTAKES

- ✗ Leaving default allow (no policies)
- ✗ Allowing all namespaces
- ✗ Overly permissive rules
- ✗ Forgetting egress rules
- ✗ Not testing policies before production



VERIQTA PRODUCTION NOTE

Network Policies give you granular control of traffic. Deny by default, then allow only what you need.



6. COMMON DEPLOYMENT STRATEGIES

Best practices for production environments.

- ✓ Start with Default Deny in every namespace
- ✓ Apply policies at the namespace level first
- ✓ Use labels for clean and maintainable rules
- ✓ Allow only required traffic (least privilege)
- ✓ Test policies in staging before production
- ✓ Monitor and audit traffic after deployment
- ✓ Review and update policies regularly

DEPLOYMENT FLOW

1. Enable CNI policy support
2. Apply default deny
3. Add allow rules
4. Monitor and improve

KEY TAKEAWAY



Network Policies turn your Kubernetes cluster into a secure, well-isolated, and production-ready environment.



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VERIQTA TIP

Ingress is your front door. Secure it well. Encrypt everything. Authenticate users. Limit abuse. Monitor always.

19. INGRESS SECURITY

SECURE THE ENTRY POINT. PROTECT YOUR APPLICATIONS.

1. TLS TERMINATION

Terminate TLS at the ingress controller or a load balancer to offload encryption and simplify backend communication.



BEST PRACTICES

- ✓ Use TLS 1.2 or 1.3
- ✓ Disable weak ciphers and protocols
- ✓ Use strong certificates (ECDSA preferred)
- ✓ Enable HSTS and secure headers
- ✓ Automate certificate renewal (Let's Encrypt)

2. HTTPS (ENCRYPTED COMMUNICATION)

Always use HTTPS for all client connections.

GOOD: HTTPS ONLY

Redirect HTTP to HTTPS



- ✓ Force HTTPS
- ✓ Use HSTS
- ✓ Strong ciphers
- ✓ Valid certificates

EXAMPLE INGRESS ANNOTATIONS

```

nginx.ingress.kubernetes.io/ssl-redirect: "true"
nginx.ingress.kubernetes.io/force-ssl-redirect: "true"
nginx.ingress.kubernetes.io/hsts: "true"
nginx.ingress.kubernetes.io/hsts-max-age: "31536000"
nginx.ingress.kubernetes.io/hsts-include-subdomains: "true"
nginx.ingress.kubernetes.io/hsts-preload: "true"
  
```

RESULT: All traffic is encrypted in transit. No plain HTTP exposure.

3. WAF INTEGRATION

Use a Web Application Firewall (WAF) to protect against common web attacks.



- ✓ OWASP Top 10 protection
- ✓ SQL Injection
- ✓ XSS Protection
- ✓ Bot Management
- ✓ IP Reputation & Geo Blocking
- ✓ Custom Rules

AWS EXAMPLE

CloudFront + AWS WAF
Application Load Balancer
→ Ingress Controller → Pods

RESULT: WAF blocks malicious traffic before it reaches your ingress and applications.

4. AUTHENTICATION

Authenticate users before allowing access to your applications.

OIDC / OAuth2



Integrate with Identity Providers (Keycloak, Okta, Auth0, Google)

mTLS



Mutual TLS for strong service and user authentication

Basic Auth



Simple user/pass authentication (Use for internal tools only)

JWT Validation



Validate JSON Web Tokens at the ingress layer

RESULT: Only authenticated and authorized users can access your applications.

5. RATE LIMITING

Protect your applications from abuse and resource exhaustion.

- ✓ Limit requests per IP
- ✓ Limit requests per user
- ✓ Limit concurrent connections
- ✓ Burst and sustained limits
- ✓ Prevent brute force attacks
- ✓ Protect backend resources

NGINX INGRESS EXAMPLE

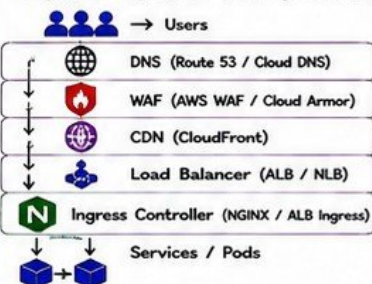
```

nginx.ingress.kubernetes.io/limit-req: "zone=one burst=20 nodelay"
nginx.ingress.kubernetes.io/limit-req-zone: "one: 10m rate=10r/s"
nginx.ingress.kubernetes.io/limit-connections: "20"
nginx.ingress.kubernetes.io/limit-rps: "10"
  
```

RESULT: Rate limiting ensures availability and prevents denial of service and abuse.

6. SECURE INGRESS DESIGN

Design your ingress for security, reliability, and performance.



SECURITY CHECKLIST

- ✓ Use HTTPS everywhere
- ✓ TLS 1.2+ only
- ✓ Strong cipher suites
- ✓ WAF enabled
- ✓ Authentication enforced
- ✓ Rate limiting enabled
- ✓ Restrict IP ranges
- ✓ Deny unnecessary paths
- ✓ Enable security headers
- ✓ Monitor and alert

RESULT: A secure ingress protects users, applications, and your entire cluster.

COMMON MISTAKES

- ✗ Exposing services without TLS
- ✗ Not redirecting HTTP to HTTPS
- ✗ Missing WAF protection
- ✗ No authentication on sensitive apps
- ✗ No rate limiting
- ✗ Overly permissive ingress rules

VERIQTA PRODUCTION NOTE



Your ingress is your internet boundary. Treat it like a firewall. Encrypt, authenticate, filter, and monitor everything that enters your cluster. Never trust traffic by default.

KEY TAKEAWAY



Secure the entry. Control the flow. Encrypt, authenticate, filter, and limit. A secure ingress means a secure application.



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20. SUPPLY CHAIN SECURITY

SECURE EVERY COMPONENT. PROTECT EVERYTHING YOU BUILD.

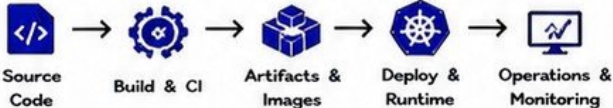


VERIQTA TIP

Trust nothing.
Verify everything.
Secure your supply chain. Secure your systems. Secure your future.

1. SUPPLY CHAIN SECURITY OVERVIEW

Your software supply chain includes every component, tool, process, and artifact used to build and run your applications.



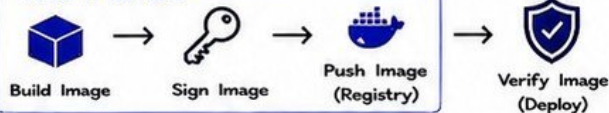
KEY GOALS

- ✓ Ensure the integrity of every component
- ✓ Prevent tampering and unauthorized changes
- ✓ Verify the origin and provenance of artifacts
- ✓ Detect and block vulnerable or malicious components
- ✓ Maintain trust from code to runtime

3. IMAGE SIGNING

Sign your container images to prove integrity and origin.

HOW IT WORKS



TOOLS

- Cosign
- Notary v2
- Docker Content Trust
- Sigstore

BENEFITS

- ✓ Verify integrity
- ✓ Verify publisher identity
- ✓ Prevent tampering
- ✓ Enable policy enforcement

RESULT: Only signed and verified images are allowed into your environments.

5. PROVENANCE

Track the origin, build process, and metadata of your artifacts.

WHAT TO CAPTURE

- ✓ Source repository
- ✓ Commit ID
- ✓ Builder environment
- ✓ Build time
- ✓ Dependencies
- ✓ Build parameters
- ✓ Signer identity

TOOLS & STANDARDS

- SLSA (Supply-chain Levels for Software Artifacts)
- in-toto
- Cosign attestations
- Tekton Chains

RESULT: Provenance shows where your artifact came from and how it was built.

2. TRUSTED IMAGES

Use trusted base images and approved sources.

BEST PRACTICES

- ✓ Use minimal, official base images
- ✓ Pull images from trusted registries
- ✓ Pin image by digest (not tag)
- ✓ Regularly scan images for vulnerabilities
- ✓ Remove unused and outdated images

EXAMPLE: USE DIGEST

❌ DO NOT USE (Mutable Tag)
nginx:latest

✅ USE (Immutable Digest)
nginx@sha256:
a3f2c1e4d5b6...

RESULT: Immutable, verified images prevent malicious updates and unexpected changes.

4. SBOM (SOFTWARE BILL OF MATERIALS)

Maintain a list of all components and dependencies.

WHAT IS AN SBOM?

A machine-readable inventory of all software components, libraries, and dependencies in your application or image.

WHY SBOM MATTERS

- ✓ Identify vulnerable components
- ✓ Improve license compliance
- ✓ Enhance transparency
- ✓ Enable faster incident response

TOOLS

- Syft
- CycloneDX
- SPDX
- Dependency-Track

RESULT: SBOM gives you visibility and control over your software components.

6. SOFTWARE INTEGRITY

Ensure code and artifacts are not altered.

PRACTICES

- ✓ Verify checksums and signatures
- ✓ Enforce signed commits
- ✓ Use immutable infrastructure
- ✓ Scan for vulnerabilities
- ✓ Detect configuration drift
- ✓ Enforce admission policies

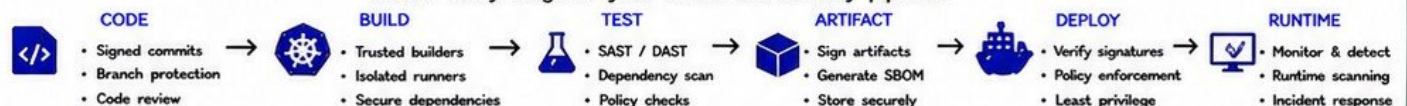
TOOLS

- Trivy
- Gripe
- Falco
- Kyverno
- OPA / Gatekeeper

RESULT: Integrity checks prevent compromised or tampered software from running.

7. ENTERPRISE PIPELINE SECURITY

Secure every stage of your CI/CD and delivery pipeline.



CODE

- Signed commits
- Branch protection
- Code review

BUILD

- Trusted builders
- Isolated runners
- Secure dependencies

TEST

- SAST / DAST
- Dependency scan
- Policy checks

ARTIFACT

- Sign artifacts
- Generate SBOM
- Store securely

DEPLOY

- Verify signatures
- Policy enforcement
- Least privilege

RUNTIME

- Monitor & detect
- Runtime scanning
- Incident response

COMMON MISTAKES

- ❌ Using untrusted base images
- ❌ Not scanning images or dependencies
- ❌ Skipping image signing and verification
- ❌ No SBOM or provenance
- ❌ Overly permissive pipeline credentials

VERIQTA PRODUCTION NOTE

A secure supply chain is not a feature. It's a foundation. Build trust, verify continuously, and never assume.

KEY TAKEAWAY

Secure the supply chain. Verify everything. Trust is earned — not assumed.



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VERIQTA TIP

A secure container image is the foundation of a secure Kubernetes workload.

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21. CONTAINER IMAGE SECURITY

SECURE IMAGES. SECURE CLUSTERS. SECURE EVERYTHING.

1. BASE IMAGE SELECTION



- ☐ Use minimal and official images from trusted sources.
- ☐ Prefer distroless or scratch images when possible.
- ☐ Avoid latest tags. Use specific version tags.
- ☐ Use LTS (Long Term Support) base images.
- ☐ Keep base images updated regularly.

EXAMPLES



GOOD
alpine:3.19
python:3.11-slim
gcr.io/distroless/base



BAD
ubuntu:latest
node:latest
centos:latest

2. IMAGE SCANNING



- ☐ Scan every image before build and before deploy.
- ☐ Use automated image scanning in CI/CD pipeline.
- ☐ Scan for OS packages, libraries, and application dependencies.
- ☐ Block critical and high severity vulnerabilities.

TOOLS



Trivy



Grype



Clair



Anchore



Snyk

3. CVE MANAGEMENT



- ☐ Track vulnerabilities (CVEs) in base images and dependencies.
- ☐ Subscribe to security advisories for your base images.
- ☐ Rebuild images frequently to pick up security fixes.
- ☐ Remove unused packages and dependencies.
- ☐ Maintain an SBOM (Software Bill of Materials).

CVE SEVERITY GUIDE

CRITICAL	HIGH	MEDIUM	LOW
9.0 - 10.0	7.0 - 8.9	4.0 - 6.9	0.1 - 3.9

Fix Critical & High immediately. Track & plan for Medium & Low.

4. REGISTRY SECURITY



- ☐ Use private container registries.
- ☐ Enforce authentication (no anonymous pulls).
- ☐ Use RBAC for registry access.
- ☐ Enable image signing and verification.
- ☐ Encrypt data in transit and at rest.
- ☐ Enable immutability for critical images.

BEST PRACTICES



Least privilege access



Short-lived credentials



Sign images (Notary/Cosign)



Enable immutability



Regular backup of registry

5. IMAGE LIFECYCLE



- ☐ Build minimal and secure images.
- ☐ Scan and validate before push.
- ☐ Store in trusted registry.
- ☐ Deploy with policy enforcement.
- ☐ Continuously monitor for new vulnerabilities.
- ☐ Rebuild and update regularly.
- ☐ Retire and remove old and unused images.

REMEMBER

Old images are invisible risks. Remove what you don't use.

6. PRODUCTION RECOMMENDATIONS



- ☐ Use minimal, trusted, and version-pinned base images.
- ☐ Scan images in CI/CD and at runtime.
- ☐ Enforce policies to block vulnerable images.
- ☐ Use private registries with strong access control.
- ☐ Sign and verify all production images.
- ☐ Continuously monitor and update images.
- ☐ Follow least privilege in everything.



FINAL SECURITY PRINCIPLE

Your containers are only as secure as the images you run. Build small, scan everything, trust nothing, verify always, and continuously improve.

A secure image today prevents a production incident tomorrow.



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22. POLICY AS CODE

AUTOMATE GOVERNANCE. ENFORCE STANDARDS. PROTECT YOUR CLUSTER.

**VERIQTA TIP**

Policies define what is allowed. Enforcement ensures what runs in your cluster meets your standards — always.

1. POLICY AS CODE OVERVIEW

Define security, compliance, and operational rules as code and enforce them automatically in your cluster.

**KEY BENEFITS**

- ✓ Consistent enforcement across all environments
- ✓ Prevent misconfigurations and security risks
- ✓ Detect and block violations early
- ✓ Version control and auditability
- ✓ Scalable governance for enterprise clusters

2. KYVERNO (KUBERNETES-NATIVE POLICY ENGINE)

Kyverno is a Kubernetes-native policy engine built for speed and simplicity.

KYVERNO CAPABILITIES

- ✓ Validate resources
- ✓ Mutate resources
- ✓ Generate resources
- ✓ Verify images
- ✓ Enforce best practices
- ✓ Audit and report
- ✓ Prevents misconfigurations

KYVERNO POLICY EXAMPLE

```
apiVersion: kyverno.io/v1
kind: ClusterPolicy
metadata:
  name: require-non-root
spec:
  validationFailureAction: enforce
  rules:
    - name: require-run-as-non-root
      match:
        any:
          - resources:
              kinds: ["Pod", "Deployment"]
            validate:
              pattern:
                spec:
                  securityContext:
                    runAsNonRoot: true
```

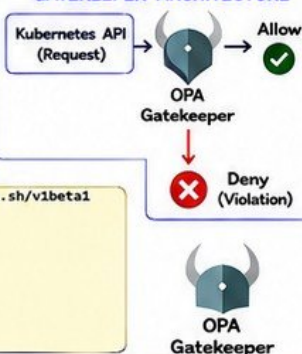
**KYVERNO POLICY TYPES**

3. OPA GATEKEEPER (CNCF POLICY ENGINE)

OPA Gatekeeper uses Rego policies to enforce governance at scale.

GATEKEEPER FEATURES

- ✓ Powerful Rego language
- ✓ Constraint-based policies
- ✓ Audit and enforcement modes
- ✓ Works cluster-wide
- ✓ Integrates with CI/CD
- ✓ Supports complex rules

GATEKEEPER ARCHITECTURE**CONSTRAINT EXAMPLE**

```
apiVersion: constraints.gatekeeper.sh/v1beta1
kind: K8sPSPAllowedHostPath
metadata:
  name: disallow-hostpath
spec:
  match:
    kinds:
      - apiGroups: ["*"]
      - kinds: ["Pod*"]
    parameters: {}
```

4. REGO FUNDAMENTALS (OPA POLICY LANGUAGE)

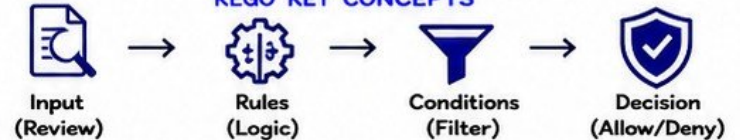
REGO BASICS

- ✓ Declarative language
- ✓ Policy-as-code
- ✓ Flexible and powerful
- ✓ Human-readable
- ✓ Works with Gatekeeper

SIMPLE REGO EXAMPLE

```
package k8srequiredlabels

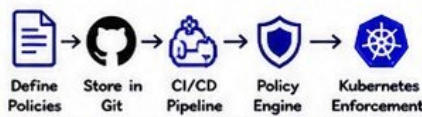
violation[msg] {
  input.review.kind.kind == "Pod"
  not input.review.object.metadata.labels.app
  msg := "Missing required label: app"
}
```

REGO KEY CONCEPTS

5. GOVERNANCE WITH POLICY AS CODE

Use policies to enforce standards and maintain compliance.

- ✓ Security Standards (Least Privilege)
- ✓ Resource Limits (CPU / Memory)
- ✓ Image Policies (Trusted Only)
- ✓ Namespace Rules (Isolation)
- ✓ Network Policies (Secure Traffic)

GOVERNANCE FLOW**WHAT YOU CAN GOVERN**

- ✓ Who can deploy what
- ✓ What images can run
- ✓ Resource usage limits
- ✓ Security context rules
- ✓ Network and access rules
- ✓ Compliance requirements



6. ENTERPRISE POLICY MANAGEMENT

Manage policies at scale with visibility, versioning, and audit.

BEST PRACTICES

- ✓ Use Git for policy versioning
- ✓ Test policies in CI/CD
- ✓ Start with audit, then enforce
- ✓ Document every policy
- ✓ Monitor and review regularly
- ✓ Keep policies simple and focused

POLICY LIFECYCLE**ENTERPRISE TOOLING ECOSYSTEM****COMMON MISTAKES**

- ✗ Not using policy as code (manual enforcement)
- ✗ Overly complex and hard-to-maintain policies
- ✗ Enforcing without testing (breaks deployments)
- ✗ Ignoring audit and monitoring
- ✗ Not versioning or documenting policies

VERIQTA PRODUCTION NOTE

Policy as Code turns your Kubernetes cluster into a governed, secure, and compliant platform. Automate decisions. Enforce standards. Protect your systems.

KEY TAKEAWAY

Policies are your guardrails. Engines enforce them. Governance protects your systems. Automation makes it scalable.

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23. VULNERABILITY MANAGEMENT

FIND EARLY. PRIORITIZE SMARTLY. PATCH SAFELY. REPORT CLEARLY.

VERIQTA TIP

Vulnerabilities don't wait for a release. Scan continuously. Fix strategically. Secure relentlessly.

1. VULNERABILITY MANAGEMENT OVERVIEW

A continuous process to identify, assess, prioritize, remediate, and report vulnerabilities across your entire environment.



KEY BENEFITS

- ✓ Reduce attack surface
- ✓ Improve compliance
- ✓ Lower risk exposure
- ✓ Faster patch cycles
- ✓ Prevent breaches
- ✓ Better visibility & accountability

3. RISK PRIORITIZATION

Focus on what matters most. Not all vulnerabilities are equal.

RISK FACTORS

- ✓ CVSS Score (Severity)
- ✓ Exploit Availability
- ✓ Asset Criticality
- ✓ Exposure (Internet/Internal)
- ✓ Business Impact
- ✓ Threat Intelligence

RISK MATRIX (Example)

IMPACT	High	Medium	High	Critical
	Medium	Low	Medium	High
	Low	Low	Low	Medium
		Low	Medium	High
		LIKELIHOOD		

RESULT: Prioritize based on risk, not just severity.

5. FALSE POSITIVES

Not every alert is real. Reduce noise and focus on real threats.

HOW TO REDUCE

- ✓ Tune scan policies
- ✓ Apply context (environment, config, reachability)
- ✓ Use suppression with justifications
- ✓ Validate before escalate
- ✓ Improve scanner accuracy

COMMON CAUSES

- ✗ Outdated vulnerability DB
- ✗ Unreachable code paths
- ✗ Misconfiguration detected as risk
- ✗ Third-party library mismatch
- ✗ Generic rules not aligned to environment

RESULT: Fewer false positives = more focus on real risks.

COMMON MISTAKES

- ✗ Scanning only at release time
- ✗ Ignoring low severity vulnerabilities
- ✗ No risk-based prioritization
- ✗ Patching without testing
- ✗ Ignoring false positives
- ✗ Poor reporting and communication

VERIQTA PRODUCTION NOTE

Vulnerability management is a continuous cycle, not a one-time task. Integrate it into your DevSecOps culture and automate everywhere possible.

KEY TAKEAWAY

Discover constantly. Prioritize wisely. Patch safely. Report clearly. Security is a process, not a project.

2. CONTINUOUS SCANNING

Automate scanning across your pipeline, infrastructure, and runtime.

WHERE TO SCAN

- </> Code (SAST)
- Dependencies (SCA)
- Containers (Image Scan)
- Infrastructure as Code
- Kubernetes (Cluster Scan)
- Runtime (CSPM / RASP)

SCAN FREQUENCY

- On Commit (Code)
- On Build (Images)
- On Push (Registries)
- Daily (Infrastructure)
- Continuous (Runtime)

TOOLS EXAMPLES



RESULT: Continuous scanning ensures early detection and faster response.

4. PATCH MANAGEMENT

Fix vulnerabilities quickly, safely, and without breaking production.

PATCH STRATEGY

- ✓ Regular patch windows
- ✓ Automated where possible
- ✓ Test before production
- ✓ Staged rollouts
- ✓ Rollback plan ready

PATCH TYPES

- Security Patches
- Bug Fixes
- Dependency Updates
- OS / Kernel Updates

BEST PRACTICES

- ✓ Maintain inventory of all assets
- ✓ Use automation (Ansible, Patch Manager, etc.)
- ✓ Validate patches in staging
- ✓ Document and track all changes
- ✓ Verify and monitor after patching

PATCH FLOW



RESULT: Timely patching reduces risk without sacrificing stability.

6. SECURITY REPORTING

Clear, actionable reports drive accountability and improvement.

REPORT CONTENT

- Executive Summary
- Risk Overview
- Top Vulnerabilities
- Remediation Status
- Trends & Metrics
- Compliance Status

REPORT TARGETS

- Engineering Teams
- Security Team
- Management / Leadership
- Audit / Compliance

RESULT: Good reporting drives action, accountability, and trust.



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VERIQTA TIP

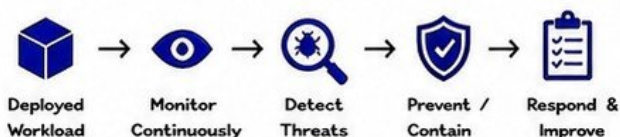
Attackers target what runs.
Monitor everything.
Detect early.
Respond fast.
Stay resilient.

24. RUNTIME SECURITY

PROTECT WHAT RUNS. DETECT WHAT STRIKES. RESPOND IN REAL TIME.

1. RUNTIME SECURITY OVERVIEW

Runtime security focuses on what happens after deployment. It detects and blocks threats in real time.



KEY BENEFITS

- ✓ Detect zero-day attacks
- ✓ Reduce mean time to detect
- ✓ Prevent lateral movement
- ✓ Protect running workloads
- ✓ Contain threats in real time
- ✓ Improve security posture

2. RUNTIME MONITORING

Collect and analyze runtime data to understand behavior.

WHAT TO MONITOR

- Process activity
- System calls
- File activity
- Network connections
- User and privilege changes
- Container and pod events
- API calls and behaviors
- Configuration changes

MONITORING SOURCES

- eBPF / Kernel Probes
- Audit Logs
- Container Runtime Logs
- System Logs
- Cloud Provider Logs
- Application Logs
- Network Flow Logs

TOOLS EXAMPLES



RESULT: Continuous visibility into everything that runs.

3. THREAT DETECTION

Detect malicious activity and policy violations in real time.

DETECTION METHODS

- Signature-based Detection
- Anomaly Detection
- Behavioral Detection
- Machine Learning
- Threat Intelligence Feeds
- Policy Violation Detection

COMMON THREATS

- Privilege Escalation
- Lateral Movement
- Container Escape
- Malware Execution
- Crypto Mining
- Data Exfiltration
- Ransomware Activity

RESULT: Early detection reduces impact and prevents spread.

4. BEHAVIORAL ANALYSIS

Understand what is normal. Detect what is not.

BEHAVIOR BASELINES

- ✓ Normal process behavior
- ✓ Normal network patterns
- ✓ Normal file access
- ✓ Normal system calls
- ✓ Normal user activities

ANOMALY EXAMPLES

- ⚠ Unexpected process spawn
- ⚠ Unusual network destinations
- ⚠ Rare system call usage
- ⚠ Unusual file modifications
- ⚠ Privilege misuse
- ⚠ Access outside baseline

RESULT: Behavioral analysis catches unknown and zero-day threats.

5. INCIDENT RESPONSE

Respond quickly. Contain. Eradicate. Recover. Learn.



RESPONSE BEST PRACTICES

- ✓ Automate response playbooks
- ✓ Use least privilege access
- ✓ Isolate affected workloads
- ✓ Preserve evidence
- ✓ Communicate clearly
- ✓ Document everything
- ✓ Review and improve

RESULT: Fast, structured response minimizes damage.

6. RUNTIME PROTECTION

Prevent attacks in real time and enforce security policies.

PROTECTION CONTROLS

- Process / Execution Control
- File Integrity Monitoring
- Network Connection Control
- System Call Filtering
- Memory Protection
- Privilege and Access Control
- Secrets and Data Protection

ENFORCEMENT EXAMPLES

- Block suspicious processes
- Deny risky system calls
- Restrict network egress
- Enforce read-only filesystems
- Drop capabilities
- Block container escapes
- Protect sensitive data

RESULT: Runtime protection stops attacks, not just detects them.

COMMON MISTAKES

- ✗ Monitoring only infrastructure
- ✗ Ignoring container and app behavior
- ✗ No runtime threat detection
- ✗ Slow or unclear incident response
- ✗ Too many alerts, no prioritization

VERIQTA PRODUCTION NOTE



Runtime security is your last line of defense. You can't stop what you can't see. Monitor everything. Detect early. Respond fast. Protect always.

KEY TAKEAWAY



Security doesn't end at deployment. Protect what runs, not just what's built. Runtime security is continuous, intelligent, and essential.

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VERIQTA TIP

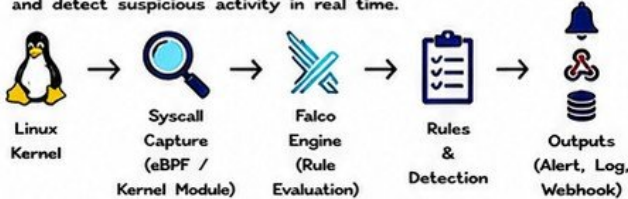
Falco gives you X-ray vision into your runtime. Detect threats before they become incidents.

25. FALCO FOR THREAT DETECTION

SEE EVERYTHING. DETECT EARLY. RESPOND FASTER.

1. FALCO ARCHITECTURE

Falco uses the Linux kernel to capture system calls and detect suspicious activity in real time.



KEY BENEFITS

- ✓ Kernel-level visibility
- ✓ Low overhead
- ✓ Real-time detection
- ✓ No code changes
- ✓ Detects known & unknown threat behaviors

2. DETECTION RULES

Falco rules define what to detect and how to alert.

RULE ANATOMY (Example)

```
- rule: Detect Shell in Container
desc: Detect a shell spawned inside a container
condition: container and proc.name in (bash, sh, zsh, ksh)
output: Shell spawned in container (%container.id) by user (%user.name) (command=%proc.cmdline)
priority: WARNING
tags: ["container", "execution", "shell"]
```

RULE ELEMENTS

- rule: Name of the rule
- desc: Description
- condition: When rule triggers
- output: Alert message
- priority: Severity level
- tags: Categories for filtering

BEST PRACTICES

- ✓ Start with Falco default rules
- ✓ Review and tune for your environment
- ✓ Reduce noise, increase signal
- ✓ Test rules in staging first

RESULT: Well-written rules catch threats with high accuracy.

3. EVENT MONITORING

Falco monitors system activity as events in real time.

WHAT FALCO MONITORS

- Process execution
- File operations
- Network connections
- User and group changes
- Container lifecycle
- Privilege escalations
- System calls

EXAMPLE EVENT FLOW



SAMPLE ALERT (JSON)

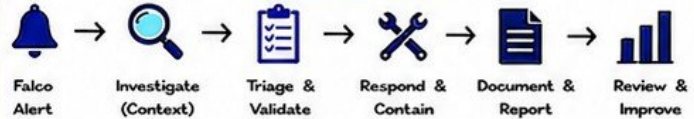
```
{
  "rule": "Detect Shell in Container",
  "priority": "WARNING",
  "output": "Shell spawned in container 1a2b3c",
  "time": "2025-07-30T10:15:25Z",
  "user": "root",
  "proc": {
    "name": "bash",
    "pid": 1234,
    "container": {
      "id": "1a2b3c",
      "name": "webapp"
    }
  }
}
```

RESULT: Continuous monitoring gives you full runtime visibility.

4. INCIDENT WORKFLOWS

From alert to action. Falco fits into your incident response process.

INCIDENT WORKFLOW



WORKFLOW DETAILS

- Alert received from Falco (UI, Slack, PagerDuty, etc.)
- Investigate: Who, What, When, Where, How
- Validate if it's true positive or noise
- Contain threat: kill process, isolate workload, revoke access
- Document incident: timeline, actions, impact
- Review rules and controls, improve detection

RESULT: A clear workflow reduces response time and impact.

5. PRODUCTION DEPLOYMENT

Deploy Falco for reliability, performance, and scale.

DEPLOYMENT OPTIONS

- DaemonSet (Kubernetes)
- Helm Chart (Recommended)
- Falco Sidekick (UI & Mgmt)
- Falco Talon (Cloud Managed)
- Custom Kernel Module

PRODUCTION CHECKLIST

- ✓ Use Falco stable version
- ✓ Enable Falco default rules
- ✓ Configure outputs (Webhook, Slack, SIEM)
- ✓ Tune rules to reduce noise
- ✓ Enable persistence (File / DB / Kafka)
- ✓ Monitor Falco health & performance
- ✓ Backup configs and rule changes

RESULT: A hardened deployment ensures Falco is always watching.

6. INTEGRATION ECOSYSTEM

Falco integrates with your tools and workflows.

INTEGRATION TARGETS

- SIEM (Splunk, Elastic, Sentinel)
- SOAR (Cortex, Shuffle, Swimlane)
- ChatOps (Slack, Teams)
- Alerting (PagerDuty, Opsgenie)
- Logging (Loki, Elasticsearch)
- Ticketing (Jira, ServiceNow)

OUTPUT METHODS

- Webhook (JSON)
- Kafka / NATS
- File / Syslog
- gRPC (Talon)
- Falco Sidekick (UI)

RESULT: Connect Falco to the tools you already use.

COMMON MISTAKES

- ✗ Ignoring default rules
- ✗ Not tuning rules (too much noise)
- ✗ No alert routing or escalation
- ✗ Running outdated Falco version
- ✗ Not monitoring Falco itself

VERIQTA PRODUCTION NOTE



Falco is your runtime defense sensor. It sees what traditional tools miss. Deploy it everywhere. Tune it smartly. Detect early. Respond faster. Protect always.

KEY TAKEAWAY



Falco brings kernel-level visibility and real-time threat detection. See it. Detect it. Stop it. That's runtime security.



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VERIQTA TIP

Logs are your truth.
Audit logs are your
evidence.
Secure them like
production data.

26. LOGGING AND AUDIT SECURITY

YOU CAN'T PROTECT WHAT YOU DON'T LOG. LOG EVERYTHING. AUDIT ALWAYS.

1. KUBERNETES AUDIT LOGS

Kubernetes audit logs capture API requests and system activity for visibility and compliance.



AUDIT EVENT EXAMPLES

- ✓ User authentication
- ✓ Resource changes
- ✓ Secret access
- ✓ RBAC changes
- ✓ Pod/Container operations
- ✓ Network policy updates

BEST PRACTICES

- ✓ Enable audit logging on all clusters
- ✓ Use a restrictive audit policy
- ✓ Log RequestResponse for sensitive resources
- ✓ Store logs securely
- ✓ Review logs regularly
- ✓ Retain logs for compliance

RESULT: Full visibility into API activity and user actions.

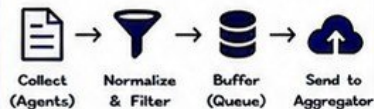
3. EVENT COLLECTION

Collect cluster events and system logs for complete context.

EVENT SOURCES

- Kubernetes Events
- API Audit Logs
- Controller Manager Logs
- Scheduler Logs
- Kubelet Logs
- Container Runtime Logs
- System Logs (Nodes)
- Application Logs

COLLECTION FLOW



COLLECTION TOOLS

fluentd fluentbit vector filebeat

RESULT: Centralized, structured, and reliable log collection.

5. CENTRALIZED LOGGING

Aggregate, index, search, and visualize logs at scale.

ARCHITECTURE



PLATFORM EXAMPLES

- Elastic Stack (ELK)
- Grafana Loki
- Splunk
- Graylog
- OpenSearch
- Azure Monitor / Sentinel
- AWS OpenSearch Service

RESULT: One place to search, analyze, and alert on everything.

COMMON MISTAKES

- ✗ Audit logging not enabled
- ✗ Overly permissive audit policy
- ✗ Storing logs on the same cluster
- ✗ No log integrity protection
- ✗ Not reviewing logs regularly

VERIQTA PRODUCTION NOTE



If it's not logged, it didn't happen.
Audit everything. Protect your logs.
Centralize, secure, and analyze.
Logs today prevent incidents tomorrow.

KEY TAKEAWAY



Logging and audit security provides visibility, accountability, and evidence. Secure your logs as seriously as your workloads.

2. API AUDITING

Audit the Kubernetes API to detect abnormal and malicious activity.

WHAT TO AUDIT

- User & ServiceAccount actions
- Admin operations
- RBAC changes
- Pod & Deployment changes
- Secrets & ConfigMaps access
- Node & Cluster changes
- Network policy modifications
- API failures & errors

AUDIT POLICY LEVELS

None	No logging
Metadata	Log metadata
Request	Log request bodies
RequestResponse	Log request + response

AUDIT POLICY EXAMPLE (YAML)

```

apiVersion: audit.k8s.io/v1
kind: Policy
rules:
- level: RequestResponse
  resources:
  - group: ""
    resources: ["pods", "secrets", "configmaps"]
- level: Metadata
  users: ["system:serviceaccount:kube-system:*"]
  
```

KEY POINTS

- ✓ Log what matters
- ✓ Balance verbosity and performance
- ✓ Protect sensitive request bodies

RESULT: API auditing helps detect threats and meet compliance.

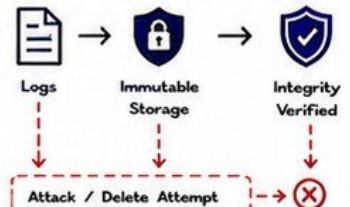
4. LOG INTEGRITY

Protect logs from tampering, deletion, and unauthorized access.

INTEGRITY CONTROLS

- Write-once storage (WORM)
- Immutable storage buckets
- Log encryption in transit & at rest
- Access control (IAM/RBAC)
- Digital signatures / Hashing
- Regular integrity verification
- Time synchronization (NTP)

TAMPER PROTECTION EXAMPLE



RESULT: Log integrity ensures trust and compliance.

6. SECURITY REPORTING

Turn logs and audits into actionable security insights.

REPORT EXAMPLES

- User activity reports
- Admin action reports
- Failed authentication reports
- RBAC change reports
- Secret access reports
- Compliance audit reports
- Threat detection reports

REPORTING CAPABILITIES

- Dashboards & visualizations
- Real-time alerts
- Scheduled reports
- Anomaly detection
- SIEM integration
- Export for compliance

RESULT: Actionable reports improve security and accountability.



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VERIQT TIP

Monitoring is your early warning system. Secure your tools, protect your data, and trust your alerts.

27. MONITORING SECURITY

SEE IT. SECURE IT. ALERT FAST. RESPOND SMART.

1. MONITORING SECURITY OVERVIEW

Secure your monitoring stack and protect your observability data. Visibility without security is a risk.



KEY BENEFITS

- ✓ Detect threats earlier
- ✓ Improve response time
- ✓ Protect monitoring data
- ✓ Ensure tool integrity
- ✓ Reduce blind spots
- ✓ Increase operational trust

RESULT: Secure monitoring delivers trusted visibility and faster response.



3. GRAFTANA SECURITY

Harden Grafana to protect dashboards and data.

SECURITY CONTROLS

- Use strong authentication (OAuth2 / SAML / OIDC)
- Enforce role-based access
- Use organization isolation
- Enable HTTPS (TLS)
- Secure data source credentials
- Disable anonymous access
- Keep Grafana updated

BEST PRACTICES

- ✓ Least privilege access
- ✓ Restrict admin users
- ✓ Use secrets management
- ✓ Audit user activity
- ✓ Backup dashboards and configs

RESULT: Secure Grafana protects insights and prevents exposure.



5. SECURITY DASHBOARDS

Visualize what matters. Detect what others miss.

ESSENTIAL SECURITY DASHBOARDS

- Cluster Security Overview
- Pod / Workload Security
- Network Security
- Identity & Access
- Compliance & Audit
- Threat Detection
- Vulnerability Summary

DASHBOARD EXAMPLE WIDGETS



RESULT: Dashboards turn data into action and awareness.



2. PROMETHEUS SECURITY

Secure your Prometheus deployment and data.

SECURITY CONTROLS

- Enable authentication (Basic Auth / OAuth2 / OIDC)
- Restrict access with RBAC
- NetworkPolicy to limit access
- Encrypt in transit (TLS)
- Encrypt at rest
- Secure remote write endpoints
- Hide sensitive metrics

BEST PRACTICES

- ✓ Expose only via internal network
- ✓ Use TLS everywhere
- ✓ Disable unused features
- ✓ Regularly rotate tokens
- ✓ Limit retention of sensitive data

PROMETHEUS YAML EXAMPLE (AUTH + TLS)

```
web:  
  basic_auth_users:  
    admin: $2y$10$hashed_password_here  
  tls_config:  
    cert_file: /etc/prometheus/tls.crt  
    key_file: /etc/prometheus/tls.key
```

KEY POINTS

- ✓ Authenticate
- ✓ Authorize
- ✓ Encrypt
- ✓ Restrict
- ✓ Monitor access

RESULT: Secure Prometheus protects metrics and prevents misuse.



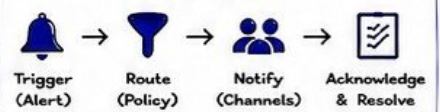
4. ALERTING

Alert on the right things, at the right time, to the right people.

ALERTING BEST PRACTICES

- ✓ Alert on symptoms not noise
- ✓ Define clear thresholds
- ✓ Use severity levels
- ✓ Route alerts by priority
- ✓ Avoid alert fatigue
- ✓ Regularly review alerts
- ✓ Test your alerts

ALERT FLOW



ALERT CHANNELS



RESULT: Smart alerting reduces noise and accelerates response.



6. OPERATIONAL VISIBILITY

Full-stack visibility across infrastructure, workloads, and users.

WHAT TO MONITOR

- Infrastructure (Nodes, CPU, Memory)
- Workloads (Pods, Deployments)
- Network (Flows, Latency, Errors)
- Security (Events, Threats, Alerts)
- Applications (APIs, Services, Logs)
- Users & Access (IAM, Actions)

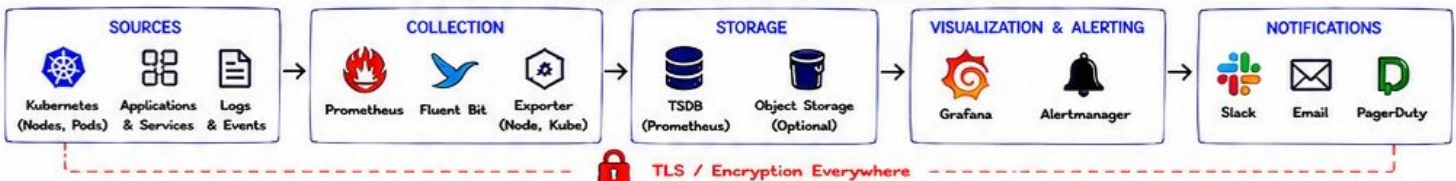
GOLDEN SIGNALS + SECURITY

- Latency
- Traffic
- Errors
- Saturation
- Security Events
- User Activity

RESULT: Complete visibility = better decisions, faster response.



7. SECURE MONITORING STACK REFERENCE ARCHITECTURE



COMMON MISTAKES

- ✗ Exposing monitoring tools publicly
- ✗ Using weak or no authentication
- ✗ Storing credentials in dashboards
- ✗ Ignoring alert fatigue
- ✗ Not backing up dashboards & alerts



VERIQT PRODUCTION NOTE

Monitoring security is not just about uptime. It's about trust, visibility, and speed. Secure your stack, tune your alerts, and always verify what matters.



KEY TAKEAWAY

If you can't see it, you can't secure it. Secure your monitoring stack, trust your data, and respond fast. Visibility is your superpower.

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VERIQTA TIP

Multi-tenancy is not just about isolation. It's about preventing one tenant from affecting another.

28. MULTI-TENANCY SECURITY

STRONG ISOLATION. CLEAR BOUNDARIES. SECURE BY DESIGN.

1. MULTI-TENANCY OVERVIEW

Run multiple teams or tenants on the same cluster without compromising security or stability.



KEY BENEFITS

- ✓ Cost efficient
- ✓ Strong isolation
- ✓ Reduced risk
- ✓ Resource sharing
- ✓ Scalable guardrails
- ✓ Better visibility
- ✓ Centralized operations
- ✓ Compliance ready
- ✓ Predictable usage

RESULT: Secure tenants. Happy teams. Stable platform.



3. RESOURCE QUOTAS

Prevent noisy neighbor problems and resource exhaustion.

QUOTA CONTROLS

- ⚙️ CPU limits
- 🧠 Memory limits
- 💾 Storage limits
- 👥 Pod count
- 📦 Service count
- 📊 Request / Limit ratio

SAMPLE RESOURCE QUOTA

```
apiVersion: v1
kind: ResourceQuota
metadata:
  name: tenant-a-quota
  namespace: tenant-a
spec:
  hard:
    requests.cpu: "4"
    requests.memory: 8Gi
    limits.cpu: "8"
    limits.memory: 16Gi
    pods: "50"
    persistentvolumeclaims: "10"
```

RESULT: Quotas enforce fairness and protect the cluster.



5. SHARED CLUSTER SECURITY

Protect the cluster while empowering tenants.

CLUSTER LEVEL CONTROLS

- 🔒 Private cluster endpoints
- 👤 RBAC least privilege
- 🛡️ Pod Security Standards
- 🏢 Admission Controllers
- 🔍 Image scanning & signing
- 📝 Audit logging
- 🔐 Secrets encryption

TENANT SELF-SERVICE (SAFE)

- ✓ Deploy within own namespace
- ✓ Use approved images
- ✓ Use predefined templates
- ✓ Request quotas via workflow
- ✓ View own metrics and logs
- ✓ No cluster admin access

RESULT: Central platform. Decentralized safely.



COMMON MISTAKES

- ✗ Using default allow NetworkPolicies
- ✗ Too many tenants in one namespace
- ✗ No resource quotas
- ✗ Sharing Secrets across tenants
- ✗ Over-privileged service accounts
- ✗ No visibility into tenant usage



VERIQTA PRODUCTION NOTE

Multi-tenancy is a strategy, not a feature. Enforce isolation at every layer: identity, network, storage, compute, and policy. Design for scale, audit everything, and never trust implicit boundaries.

2. NAMESPACE ISOLATION

Namespaces provide the first boundary.

ISOLATION CONTROLS

- 👤 RBAC per namespace
- 🔗 NetworkPolicies between namespaces
- 🔒 Limit cross-namespace communication
- 🔑 Separate Secrets and ConfigMaps
- 👤 No default service account token
- ⚙️ Disable privilege escalation
- 📊 Distinct resource limits

BEST PRACTICES

- ✓ One tenant = one namespace
- ✓ Deny all traffic by default
- ✓ Use NetworkPolicies
- ✓ Use least-privilege RBAC
- ✓ Separate secrets per tenant
- ✓ Avoid cross-namespace access

NETWORK POLICY EXAMPLE (DENY ALL)

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny
  namespace: tenant-a
spec:
  podSelector: {}
  policyTypes: ["Ingress", "Egress"]
```

KEY POINTS

- ✓ Default deny
- ✓ Allow what's needed
- ✓ Micro-segmentation
- ✓ Reduce blast radius

RESULT: Namespaces + policies = strong isolation layer.



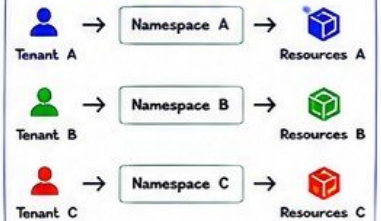
4. TENANT BOUNDARIES

Define clear boundaries for identity, network, storage and data.

BOUNDARY AREAS

- 👤 Identity boundary (RBAC / Groups)
- 🔗 Network boundary (NetworkPolicies)
- 💾 Storage boundary (PVC / StorageClass)
- 🔒 Data boundary (Secrets / Encryption)
- ⚙️ Compute boundary (Quotas / Limits)
- 🛡️ Policy boundary (OPA / Kyverno)

BOUNDARY EXAMPLES



RESULT: Clear boundaries = fewer risks and incidents.



6. ENTERPRISE MULTI-TENANCY DESIGN

A secure, scalable, and observable multi-tenant architecture.



RESULT: Secure design. Scalable tenants. Enterprise ready.



KEY TAKEAWAY

Isolation enables innovation. Guardrails enable scale. Observability enables trust. Secure the platform. Empower the tenants.

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VERIQTA TIP

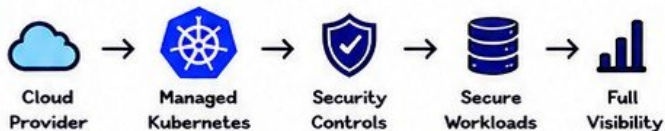
Managed Kubernetes takes care of the control plane, but security is still your responsibility.

29. CLOUD KUBERNETES SECURITY

USE MANAGED POWER. SECURE BY DESIGN. PROTECT EVERYTHING.

1. CLOUD KUBERNETES SECURITY OVERVIEW

Managed Kubernetes provides secure, scalable control planes. You secure the workloads, network, identity and data.



KEY BENEFITS

- ✓ Managed control plane
- ✓ Scalability
- ✓ Reduced ops work
- ✓ Built-in security features
- ✓ Integrated IAM
- ✓ Stronger defaults
- ✓ High availability
- ✓ Compliance support
- ✓ Faster deployment

RESULT: Enterprise-grade, secure, and reliable Kubernetes platform.

3. CLOUD IAM SECURITY

Strong identity and access management for cloud + Kubernetes.

BEST PRACTICES

- Use least privilege
- Use short-lived credentials
- Use roles, not users
- Enable audit logging
- Enforce MFA
- Use workload identities

WORKLOAD IDENTITY EXAMPLE (EKS)



```
# ServiceAccount with IAM role (EKS IRSA)
apiVersion: v1
kind: ServiceAccount
metadata:
  name: app-sa
  namespace: production
  annotations:
    eks.amazonaws.com/role-arn: arn:aws:iam::123456789012:role/app-role
```

RESULT: Secure access to cloud resources without static credentials.

5. CLOUD-NATIVE SECURITY FEATURES

Use native cloud tools for deeper protection.

AWS SECURITY STACK

- AWS WAF
- AWS Shield
- AWS Inspector
- GuardDuty
- Security Hub
- CloudTrail

AZURE SECURITY STACK

- WAF
- Defender for Cloud
- Microsoft Sentinel
- Azure Policy
- Key Vault
- Monitor

GOOGLE CLOUD STACK

- Cloud Armor
- Security Command Center
- Binary Authorization
- Cloud Audit Logs
- Secret Manager
- Anthos Policy Controller

RESULT: Layer cloud-native controls with Kubernetes security practices.

COMMON MISTAKES

- ✗ Using public clusters in production
- ✗ Overly permissive IAM policies
- ✗ No network segmentation
- ✗ Ignoring audit logging
- ✗ Forgetting to enable encryption



VERIQTA PRODUCTION NOTE

Cloud Kubernetes gives you the power. Security gives you the protection. Combine both, and you get a scalable, resilient, and enterprise-ready platform.

KEY TAKEAWAY



Managed Kubernetes is not automatically secure. Use cloud-native security features, strong IAM, private clusters, and best practices to protect your platform.

2. MANAGED KUBERNETES SERVICES

Compare the major cloud Kubernetes platforms.

AMAZON EKS



- ✓ Managed control plane
- ✓ IAM integration
- ✓ VPC native
- ✓ Private clusters
- ✓ AWS security tooling
- ✓ Fargate for serverless

AZURE AKS



- ✓ Managed control plane
- ✓ Azure AD integration
- ✓ Private clusters
- ✓ Azure Policy
- ✓ Defender for Cloud
- ✓ Virtual node scaling

GOOGLE GKE



- ✓ Managed control plane
- ✓ Workload Identity
- ✓ Private clusters
- ✓ Binary Authorization
- ✓ Shielded nodes
- ✓ Autopilot mode

QUICK COMPARISON

Feature	EKS	AKS	GKE
IAM Integration	AWS IAM	Azure AD	Workload Identity
Policy as Code	IAM + OPA	Azure Policy	Built-in Policy
Security Scanner	Inspector	Defender	Container + CA
Serverless Option	Fargate	Virtual Nodes	Autopilot

RESULT: Choose the platform that fits your enterprise and compliance needs.

4. MANAGED CONTROL PLANES

Security features provided by cloud providers.

SECURITY FEATURES

- Encrypted control plane
- RBAC enabled by default
- Automated patching
- High availability
- DDoS protection
- Network isolation
- Audit logging

OPERATION MODES



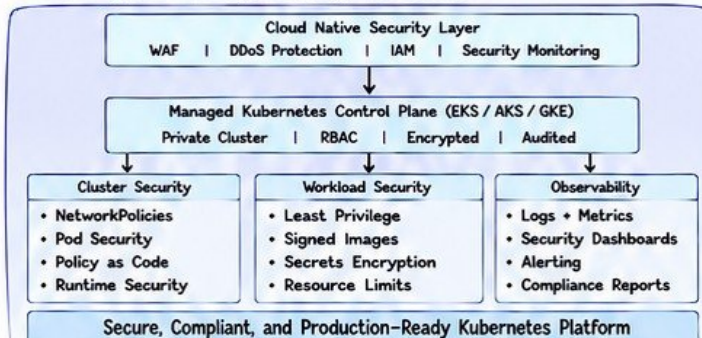
KEY POINTS

- ✓ Use private clusters in production
- ✓ Restrict API server access
- ✓ Enable audit logs
- ✓ Keep platforms updated

RESULT: Managed control planes provide strong foundation security.

6. END-TO-END CLOUD KUBERNETES SECURITY

Build a complete multi-layer security architecture.



Secure, Compliant, and Production-Ready Kubernetes Platform



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VERIQTA TIP

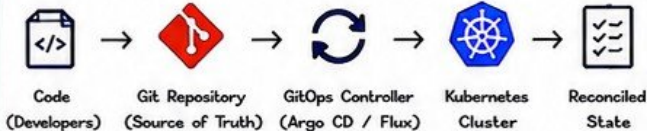
GitOps is code.
Treat your Git repositories as critical infrastructure.
Secure the source.
Secure the cluster.

30. GITOPS SECURITY

SECURE THE REPO. VERIFY THE CODE. TRUST THE DELIVERY.

1. GITOPS SECURITY OVERVIEW

GitOps uses Git as the single source of truth.
Security must protect the entire delivery chain.



GITOPS SECURITY PRINCIPLES

- ✓ Protect the Git repository
- ✓ Verify code integrity
- ✓ Least privilege everywhere
- ✓ Automate secure delivery
- ✓ Detect drift and anomalies
- ✓ Audit everything

RESULT: Secure the GitOps pipeline end-to-end.

2. GITOPS TOOLS: ARGO CD & FLUX

ARGO CD



- ✓ Declarative continuous delivery
- ✓ UI, CLI, API
- ✓ Application health monitoring
- ✓ RBAC & SSO integration
- ✓ Project-level isolation
- ✓ Audit logs & notifications

FLUX



- ✓ Kustomize / Helm native
- ✓ Git-based reconciliation
- ✓ Multi-tenancy with namespaces
- ✓ Image automation (Flux Image)
- ✓ Source verification
- ✓ Lightweight & cloud-native

COMMON BENEFITS



RESULT: GitOps tools automate delivery with security and consistency.

3. REPOSITORY PROTECTION

PROTECT YOUR REPOS

- Private repositories only
- Restrict who can push
- Require pull requests
- Enable branch protection
- Restrict force pushes
- Limit who can merge

BRANCH PROTECTION RULES (EXAMPLE)

Branch: main
Require pull request reviews
Minimum reviewers: 2
Dismiss stale reviews: true
Require status checks: true
Require signed commits: true
Restrict pushes: true
Allow force pushes: false

RESULT: Repository protection prevents unauthorized changes.

4. SIGNED COMMITS & CODE INTEGRITY

SIGN COMMITS



GPG / SSH SIGNING

```
# Generate GPG key
gpg --full-generate-key
# Sign commit
git commit -S -m "secure change"
# Verify commit
git log --show-signature
```

VERIFY EVERYTHING

- ✓ Require signed commits
- ✓ Verify author identity
- ✓ Enforce commit signature
- ✓ Block unsigned changes
- ✓ Audit commit history
- ✓ Detect tampering

RESULT: Signed commits ensure code integrity and authenticity.

5. SECURE DEPLOYMENTS

SECURE DELIVERY PRACTICES

- Use least-privilege service accounts
- Restrict namespace access
- Use signed and verified images
- Enforce policy as code
- Scan manifests & images
- Automate rollback on failure

ARGO CD / FLUX SECURITY TIPS

- ✓ Enable RBAC & SSO
- ✓ Restrict project access
- ✓ Use read-only deploy keys
- ✓ Enforce network policies
- ✓ Limit cluster admin access
- ✓ Use managed secrets (SealedSecrets)

RESULT: Secure deployments reduce risk and ensure reliability.

6. GITOPS GOVERNANCE

GOVERNANCE CONTROLS

- Define ownership of repositories
- Standardize repo structure
- Enforce code review policy
- Define deployment workflows
- Track changes and approvals
- Maintain compliance & audit logs

EXAMPLE REPO STRUCTURE

```
infrastructure/
├── clusters/
│   ├── dev/
│   ├── staging/
│   └── prod/
├── components/
├── policies/
├── docs/
└── README.md
```

RESULT: Strong governance ensures consistent, compliant delivery.

7. GITOPS SECURITY ARCHITECTURE



COMMON MISTAKES

- ✗ Exposing Git repositories
- ✗ Allowing unsigned commits
- ✗ Weak branch protection
- ✗ Over-privileged service accounts
- ✗ Ignoring drift and audit logs
- ✗ No rollback or recovery plan

VERIQTA PRODUCTION NOTE

GitOps security is not optional.
Secure the repository, verify the code, enforce policy, and monitor everything.
Trust is earned in Git, not assumed in the cluster.

KEY TAKEAWAY

Git is your production control plane.
Protect repositories and identities.
Verify every change.
Automate secure deployments.
Govern for scale and compliance.



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31. CI/CD SECURITY FOR KUBERNETES

SECURE THE PIPELINE. PROTECT THE CLUSTER. DELIVER WITH CONFIDENCE.

VERIQTA TIP

A secure pipeline is the first line of defense for your Kubernetes workloads.

1. CI/CD SECURITY OVERVIEW

Every stage of the pipeline can introduce risk. Secure the code, the build, and the delivery.



SECURE PIPELINE PRINCIPLES

- ✓ Secure by default
- ✓ Least privilege access
- ✓ Verify everything
- ✓ Automate security checks
- ✓ Protect secrets & artifacts
- ✓ Audit and improve continuously

RESULT: Secure pipelines reduce risk and ensure reliable delivery.

3. SECRET PROTECTION

Protect secrets at rest, in transit, and in use.

BEST PRACTICES

- Store secrets in vaults
- Use short-lived tokens
- Inject secrets at runtime
- Never hardcode secrets
- Restrict secret access (RBAC)
- Encrypt secrets in transit & at rest
- Rotate secrets regularly

TOOLS & INTEGRATIONS

- HashiCorp Vault
- AWS Secrets Manager
- Azure Key Vault
- Bitnami Sealed Secrets
- External Secrets Operator

RESULT: Secrets remain confidential and accessible only when needed.

5. DEPLOYMENT APPROVALS

Use policy and human approvals for sensitive environments.

APPROVAL STRATEGIES

- Manual approvals for prod
- Environment protection rules
- Change advisory automation
- Time-based deployment windows
- Multi-level approvals (SoD)
- ChatOps approval (Slack / Teams)
- Rollback plan before approval

EXAMPLE: GITLAB ENV PROTECTION

Environment: production

- Required approvers: 2
- Allowed to deploy: Maintainers
- Prevent merge requests: Yes
- Wait for successful scans: Yes

RESULT: Approved changes reduce risk and increase accountability.

2. SECURE PIPELINES

Harden every stage of your CI/CD workflow.

SECURE PIPELINE CONTROLS

- Use least-privilege service accounts
- Isolate build environments
- Pin pipeline dependencies
- Disable shell access in runners
- Enforce branch protections
- Require code review & approvals
- Audit pipeline configuration changes
- Use signed commits & tags

EXAMPLE: GITLAB CI SECURE STAGE

```
deploy:
  stage: deploy
  image: bitnami/kubectl:latest
  script:
    - kubectl apply -f manifests/
  environment: production
  only:
    - main
  when: manual # Requires approval
```

PIPELINE FLOW (SECURE)



KEY POINTS

- ✓ No direct deploy on commit
- ✓ Manual approval for prod
- ✓ Isolated and reproducible builds
- ✓ Scan before you deploy
- ✓ Audit all pipeline activity

RESULT: Secure pipelines prevent insecure code from reaching production.

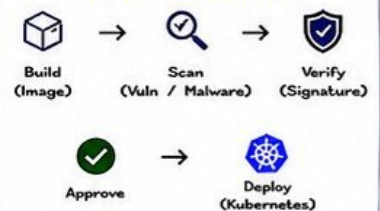
4. IMAGE VALIDATION

Ensure only trusted images are built and deployed.

IMAGE SECURITY CONTROLS

- Scan images for vulnerabilities
- Enforce trusted registries
- Verify image signatures
- Use admission controls
- Block untrusted or unsigned images
- Verify base image integrity
- SBOM generation & verification

IMAGE VALIDATION FLOW



RESULT: Only trusted, verified images run in your clusters.

6. ARTIFACT INTEGRITY

Verify artifacts to ensure they are authentic and untampered.

INTEGRITY CONTROLS

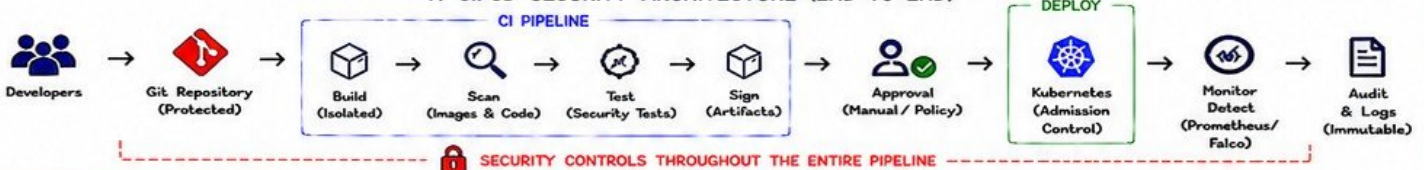
- Sign artifacts and images
- Verify checksums (SHA256)
- Use immutable artifact repositories
- Enable provenance (SLSA)
- Generate and store SBOM
- Verify dependencies & packages
- Audit artifact access

INTEGRITY FLOW



RESULT: Artifact integrity ensures authenticity and prevents tampering.

7. CI/CD SECURITY ARCHITECTURE (END-TO-END)



COMMON MISTAKES

- ✗ Hardcoding secrets in code or manifests
- ✗ Using over-privileged service accounts
- ✗ Skipping image scans
- ✗ Deploying without approvals
- ✗ Not verifying image signatures
- ✗ Storing artifacts in mutable locations

VERIQTA PRODUCTION NOTE

A secure pipeline is not optional — it's foundational. Automate security, verify everything, and enforce policies consistently. Security built into CI/CD is security that scales.

KEY TAKEAWAY

Secure the pipeline. Protect secrets and artifacts. Verify code, images, and deployments. Use approvals and governance. Trust is earned, not assumed.

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32. COMPLIANCE AND GOVERNANCE

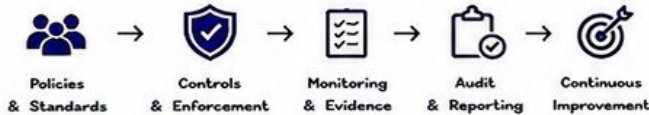
BUILD TRUST. STAY COMPLIANT. OPERATE WITH CONFIDENCE.

VERIQTA TIP

Compliance is not a checkbox. It is a continuous practice that protects your business and users.

1. COMPLIANCE & GOVERNANCE OVERVIEW

Kubernetes environments must meet organizational, industry, and regulatory requirements. Strong governance ensures security, accountability, and audit readiness.



GOVERNANCE PRINCIPLES

- ✓ Accountability
- ✓ Transparency
- ✓ Least Privilege
- ✓ Consistency
- ✓ Risk-based
- ✓ Traceability
- ✓ Continuous Monitoring
- ✓ Evidence Collection
- ✓ Compliance by Design

RESULT: Aligned, secure, and auditable Kubernetes operations.

3. NIST CYBERSECURITY FRAMEWORK (CSF)

Use NIST CSF to manage and reduce cybersecurity risk.

CORE FUNCTIONS

- Identify Understand your environment
- Protect Implement safeguards
- Detect Identify anomalies
- Respond Take action on incidents
- Recover Restore capabilities

KUBERNETES APPLICATION

- ✓ Asset inventory & classification
- ✓ Access control & identity
- ✓ Network segmentation
- ✓ Threat detection & monitoring
- ✓ Incident response playbooks
- ✓ Backup & disaster recovery
- ✓ Lessons learned & improvement

RESULT: Structured cybersecurity risk management.

5. SOC 2 COMPLIANCE

Demonstrate trust through security, availability, and confidentiality.

5 TRUST SERVICE CRITERIA

- Security
- Availability
- Confidentiality
- Processing Integrity
- Privacy

KUBERNETES IMPLEMENTATION

- ✓ Access management (RBAC, IAM)
- ✓ High availability architecture
- ✓ Data classification & encryption
- ✓ Change management & reviews
- ✓ Logging, monitoring & retention
- ✓ Incident response & communication

RESULT: Trust and confidence for customers and partners.

2. CIS KUBERNETES BENCHMARK

A security benchmark for Kubernetes. Follow the Controls.

CONTROL CATEGORIES

- Control Plane Security
- Worker Node Security
- Policy Enforcement
- Pod Security Standards
- Network Security
- Secrets Management
- Logging & Monitoring

KEY PRACTICES

- ✓ Harden API server
- ✓ Enable RBAC
- ✓ Use Pod Security Standards
- ✓ Restrict privileged containers
- ✓ Encrypt secrets
- ✓ Configure audit logging
- ✓ Monitor and alert

TOOLS



kube-bench



kube-hunter



polaris



Falco



OPA/Gatekeeper

RESULT: Secure baseline aligned to CIS best practices.

4. ISO 27001 COMPLIANCE

Information Security Management System (ISMS) for Kubernetes.

KEY ISO 27001 DOMAINS

- A.5 Information Security Policies
- A.6 Organization of Information Security
- A.9 Access Control
- A.12 Operations Security
- A.13 Communications Security
- A.14 System Acquisition & Development
- A.18 Compliance

KUBERNETES CONTROLS

- ✓ Document security policies
- ✓ Enforce RBAC & least privilege
- ✓ Protect secrets & data
- ✓ Maintain system hardening
- ✓ Encrypt data in transit/at rest
- ✓ Secure SDLC & images
- ✓ Monitor & audit continuously

RESULT: Strong ISMS that meets global standards.

6. PCI DSS COMPLIANCE

Protect cardholder data in Kubernetes workloads.

PCI DSS OBJECTIVES

- Build and maintain secure networks
- Protect cardholder data
- Maintain vulnerability management
- Implement strong access control
- Monitor and test networks
- Maintain information security policy

KUBERNETES CONTROLS

- ✓ Network segmentation
- ✓ Encrypt cardholder data
- ✓ Restrict access to data
- ✓ Patch nodes & images
- ✓ Monitor and log all access
- ✓ Use secure configurations

RESULT: Secure environments for cardholder data.

7. REGULATORY ALIGNMENT MATRIX

REQUIREMENT AREA	CIS BENCHMARK	NIST CSF	ISO 27001	SOC 2	PCI DSS	KEY CONTROLS
Access Control & IAM	✓	✓	✓	✓	✓	RBAC, least privilege, MFA, IAM integration
Network Security	✓	✓	✓	✓	✓	Network Policies, segmentation, firewall, encryption
Data Protection	✓	✓	✓	✓	✓	Encryption, secrets management, data classification
Logging & Monitoring	✓	✓	✓	✓	✓	Audit logs, monitoring, alerting, log retention
Vulnerability Management	✓	✓	✓	✓	✓	Image scanning, patching, CIS benchmark
Incident Response	✓	✓	✓	✓	✓	Playbooks, detection, response, recovery
Policy & Governance	✓	✓	✓	✓	✓	Policies, documentation, training, reviews

COMMON MISTAKES

- ✗ Treating compliance as a one-time task
- ✗ Not mapping controls to frameworks
- ✗ Missing evidence & audit trails
- ✗ Over-privileged access
- ✗ Ignoring continuous monitoring

VERIQTA PRODUCTION NOTE



Compliance is built, not bought.
Automate controls, collect evidence,
and review continuously. Strong governance
turns compliance into a competitive advantage.

KEY TAKEAWAY



Map controls to frameworks.
Automate enforcement and monitoring.
Document everything.
Review, improve, repeat.
Compliance is a journey, not a destination.

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VERIQTA TIP

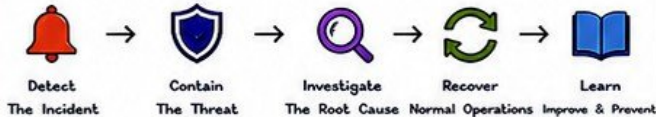
A strong incident response plan turns chaos into control and minimizes damage.

33. INCIDENT RESPONSE

DETECT FAST. RESPOND SMART. RECOVER STRONGER.

1. INCIDENT RESPONSE OVERVIEW

A structured process to handle security incidents in your Kubernetes environment.



INCIDENT RESPONSE PRINCIPLES

- ✓ Prepare before incidents happen
- ✓ Act fast and decisively
- ✓ Communicate clearly
- ✓ Minimize impact
- ✓ Document everything
- ✓ Improve continuously

RESULT: Reduce impact, speed recovery, and strengthen security.

3. CONTAINMENT

Stop the threat from spreading and limit damage.

IMMEDIATE ACTIONS

- ✓ Isolate affected pods/nodes
- ✓ Block malicious IPs / traffic
- ✓ Revoke compromised credentials
- ✓ Disable affected user/service accounts
- ✓ Enforce NetworkPolicies
- ✓ Scale down or cordon nodes

CONTAINMENT STRATEGIES

- ✓ Quarantine namespace
- ✓ Limit egress traffic
- ✓ Rotate secrets & keys
- ✓ Snapshot affected workloads
- ✓ Apply emergency RBAC restrictions
- ✓ Enable read-only mode if needed

RESULT: Contain the threat and prevent further impact.

5. RECOVERY

Restore normal operations securely.

RECOVERY ACTIONS

- ✓ Remove malicious workloads
- ✓ Rebuild or clean affected pods
- ✓ Restore from backups (if needed)
- ✓ Patch vulnerabilities
- ✓ Validate system integrity
- ✓ Re-enable services gradually

RECOVERY CHECKLIST

- ✓ Confirm threat is fully removed
- ✓ Validate RBAC & access controls
- ✓ Check secrets & certificates
- ✓ Monitor for reoccurrence
- ✓ Test critical functionality
- ✓ Communicate recovery status

RESULT: Restore services safely and validate integrity.

2. DETECTING INCIDENTS

Identify threats early using multiple signals.

DETECTION METHODS

- Falco alerts & rules
- Kubernetes audit logs
- Prometheus alerting
- SIEM / Log correlation
- Network anomaly detection
- Vulnerability scanners
- User & access anomalies

WHAT TO WATCH FOR

- Unexpected pod creations
- Privileged containers
- Unauthorized access
- Sensitive data access
- High resource usage
- Failed login attempts
- Malicious network traffic

TOOLS & INTEGRATIONS



RESULT: Detect incidents early and reduce dwell time.

4. INVESTIGATION

Find the root cause and understand the full impact.

INVESTIGATION STEPS

- Collect logs & events
- Analyze timelines
- Identify attack vector
- Determine scope
- Assess data impact
- Preserve evidence
- Document findings

KEY QUESTIONS

- ? How did it happen?
- ? What was accessed or changed?
- ? Which users or services were involved?
- ? What data was affected?
- ? How far did the attacker move?
- ? What vulnerabilities were exploited?
- ? What needs to be fixed?

RESULT: Understand the incident and its full impact.

6. LESSONS LEARNED

Improve your defenses and prevent repeat incidents.

POST-INCIDENT REVIEW

- ✓ What happened and why?
- ✓ What worked well?
- ✓ What didn't work?
- ✓ What can be improved?
- ✓ Update runbooks & playbooks
- ✓ Share knowledge with team

IMPROVEMENT ACTIONS

- ✓ Update security policies
- ✓ Improve monitoring & alerts
- ✓ Enhance detection rules
- ✓ Conduct training & drills
- ✓ Fix root cause vulnerabilities
- ✓ Strengthen access controls

RESULT: Learn from incidents and become more resilient.

7. INCIDENT RESPONSE PLAYBOOK WORKFLOW



PLAYBOOK STORED IN GIT

Version controlled
Reviewed & tested
Always up to date

COMMON INCIDENT TYPES

- ✗ Malicious pod / container
- ✗ Privilege escalation
- ✗ Credential compromise
- ✗ Data exfiltration
- ✗ Ransomware
- ✗ Misconfiguration exposure

VERIQTA PRODUCTION NOTE

Incident response is a team sport. Prepare, practice, and document your processes. The faster you respond, the smaller the damage.

KEY TAKEAWAY

Detect early.
Act fast.
Investigate thoroughly.
Recover safely.
Learn and improve continuously.



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VERIQTA TIP

Backups you never test are just assumptions. Test restores, document runbooks, and protect your backups.

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34. DISASTER RECOVERY AND BACKUP SECURITY

PREPARE TODAY. RESTORE TOMORROW. SURVIVE ANYTHING.

1. DISASTER RECOVERY OVERVIEW

A resilient Kubernetes environment can recover from any failure scenario.



DR PRINCIPLES

- ✓ Assume failures will happen
- ✓ Have tested, automated backups
- ✓ Minimize Recovery Time Objective (RTO)
- ✓ Document and practice runbooks
- ✓ Minimize Recovery Time Objective (RTO)
- ✓ Continuously improve resilience

RESULT: Resilient, tested, and reliable recovery capability.

3. RESTORE VALIDATION

Test restores to ensure backups actually work.

RESTORE VALIDATION CHECKLIST

- ✓ Regular restore drills
- ✓ Restore in isolated environment
- ✓ Validate data integrity
- ✓ Validate application functionality
- ✓ Test different scenarios (partial/full)
- ✓ Measure RTO and RPO
- ✓ Document results and issues
- ✓ Automate restore testing when possible

RESTORE SCENARIOS

- Single Pod / Deployment
- Namespace
- Cluster Add-ons
- Full Cluster (All workloads)
- Cross-region restore

RESULT: Verified restores ensure confidence during real disasters.

5. KEY MANAGEMENT

Secure your encryption keys. Protect your kingdom.

KEY MANAGEMENT PRACTICES

- ✓ Use KMS / Cloud KMS
- ✓ Enable automatic key rotation
- ✓ Limit key usage permissions
- ✓ Separate keys per environment
- ✓ Audit key usage
- ✓ Protect service account keys
- ✓ Never store keys in code

KEY MANAGEMENT TOOLS



RESULT: Keys are secured, rotated, and centrally managed.

2. BACKUP PROTECTION

Protect backups at rest, in transit, and in storage.

BACKUP PROTECTION CONTROLS

- ✓ Encrypt backups (AES-256)
- ✓ Encrypt in transit (TLS 1.2+)
- ✓ Store backups in private buckets
- ✓ Enable bucket versioning
- ✓ Enable object lock / immutability
- ✓ Restrict access with IAM policies
- ✓ Use KMS for encryption keys
- ✓ Audit backup access and changes

BEST PRACTICES

- ✓ Separate backup account/project
- ✓ Least privilege access only
- ✓ MFA for backup access
- ✓ Regular access key rotation
- ✓ Monitor backup storage
- ✓ Alert on backup failures
- ✓ Validate backup integrity
- ✓ Protect backup metadata

COMMON BACKUP TOOLS



RESULT: Backups are protected, encrypted, and tamper-resistant.

4. ETCD RECOVERY

etcd is critical. Protect it like your data depends on it (it does).

ETCD BACKUP STRATEGIES

- ✓ Automated etcd snapshots
- ✓ Regular snapshot schedule
- ✓ Store off-cluster / off-region
- ✓ Encrypt snapshots
- ✓ Keep multiple versions
- ✓ Verify snapshot integrity
- ✓ Monitor snapshot success

ETCD RECOVERY STEPS

1. Stop Kubernetes API server
2. Stop etcd service
3. Restore latest snapshot
4. Verify snapshot integrity
5. Start etcd service
6. Start Kubernetes API server
7. Validate cluster health

RESULT: etcd can be restored quickly and safely.

6. BUSINESS CONTINUITY

Keep the business running, even when disaster strikes.

BUSINESS CONTINUITY CONTROLS

- ✓ Multi-AZ and Multi-Region deployments
- ✓ Cross-region backups
- ✓ Global DNS failover
- ✓ Runbooks and escalation paths
- ✓ Communication plans
- ✓ Third-party dependency checks
- ✓ Regular DR exercises

CONTINUITY METRICS

- RTO (Recovery Time Objective)
Max acceptable downtime
- RPO (Recovery Point Objective)
Max acceptable data loss
- Availability Target
(eg: 99.9% or higher)

RESULT: Business continues with minimal disruption.

7. DISASTER RECOVERY ARCHITECTURE (EXAMPLE)



COMMON DISASTER SCENARIOS

- ✗ Region / AZ Outage
- ✗ etcd Data Loss
- ✗ Ransomware Attack
- ✗ Accidental Deletion
- ✗ Backup Corruption
- ✗ Cloud Provider Outage

VERIQTA PRODUCTION NOTE

Disaster recovery is not just about backups. It's about preparation, automation, testing, and people. The best DR plan is the one you practice.

KEY TAKEAWAY

Protect backups.
Test restores.
Secure your keys.
Document everything.
Be ready before disaster happens.



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35. PRODUCTION SECURITY TROUBLESHOOTING

FIND IT. FIX IT. SECURE IT. KEEP IT RUNNING.

VERIQTA TIP

In production, speed and accuracy save systems. Follow a method. Verify everything. Document what you do.



1. AUTHENTICATION FAILURES

Users, services, or pods cannot authenticate to the cluster.

COMMON CAUSES

- Expired / invalid tokens
- Wrong kubeconfig context
- Incorrect client certificate
- OIDC / IdP misconfiguration
- Service account token issues

TROUBLESHOOTING STEPS

- 1 Verify kubeconfig context
- 2 Check token expiration
- 3 Validate client certificate
- 4 Check OIDC / IdP configuration
- 5 Verify service account token
- 6 Review API server logs

USEFUL COMMANDS

```
kubectl config get-contexts
kubectl config use-context <ctx>
kubectl auth can-i --list
kubectl get --raw /healthz
kubectl logs -n kube-system -l component=kube-apiserver
```

RESULT: Restore valid authentication and verify access.



2. RBAC ISSUES

Users or workloads do not have the right permissions.

COMMON CAUSES

- Missing Role / ClusterRole
- Role not bound to user
- Incorrect namespace
- Wrong verb or resource
- Aggregated role not updated

TROUBLESHOOTING STEPS

- 1 Check permissions (can-i)
- 2 Verify Role / ClusterRole
- 3 Check RoleBinding / ClusterRoleBinding
- 4 Confirm namespace scope
- 5 Review API server logs
- 6 Test with impersonation

USEFUL COMMANDS

```
kubectl auth can-i <verb> <resource>
kubectl get roles -A
kubectl get rolebindings -A
kubectl get clusterroles
kubectl auth can-i --as=<user> <verb> <resource> -n <ns>
```

RESULT: Grant correct permissions and validate access.



3. NETWORK POLICY DEBUGGING

Pods cannot communicate due to network restrictions.

COMMON CAUSES

- Deny all policy blocking traffic
- Missing allow rule
- Wrong label selector
- Egress policy blocking
- CNI plugin issues

TROUBLESHOOTING STEPS

- 1 List all NetworkPolicies
- 2 Check pod labels
- 3 Describe the policy
- 4 Test connectivity (from & to)
- 5 Check CNI plugin logs
- 6 Review default deny behavior

USEFUL COMMANDS

```
kubectl get networkpolicies -A
kubectl describe networkpolicy <name> -n <ns>
kubectl get pods -l <label> -n <ns>
kubectl exec -it <pod> -- nc -zv <ip> <port>
kubectl logs -n kube-system -l k8s-app=cni-plugin
```

RESULT: Adjust NetworkPolicies and verify connectivity.



4. ADMISSION FAILURES

Pods are rejected by admission controllers or policies.

COMMON CAUSES

- PodSecurity admission
- OPA / Gatekeeper denial
- Kyverno policy violation
- Image signature / policy
- Resource limit violations

TROUBLESHOOTING STEPS

- 1 Describe the failing resource
- 2 Check events for denial reason
- 3 Review admission controller logs
- 4 Validate policy configuration
- 5 Test with --dry-run=server
- 6 Fix policy or request and retry

USEFUL COMMANDS

```
kubectl describe pod <pod> -n <ns>
kubectl get events -n <ns> --sort-by=.lastTimestamp
kubectl logs -n gatekeeper-system
kubectl logs -n kyverno
kubectl apply --dry-run=server -f <file.yaml>
```

RESULT: Satisfy admission policies and deploy successfully.



5. SECRET PROBLEMS

Secrets not mounting, wrong data, or access issues.

COMMON CAUSES

- Secret not found
- Wrong secret name
- Wrong namespace
- Incorrect key
- RBAC blocks access

TROUBLESHOOTING STEPS

- 1 Check secret exists
- 2 Verify namespace
- 3 Confirm secret keys
- 4 Check pod volume mounts
- 5 Review pod events
- 6 Check RBAC permissions

USEFUL COMMANDS

```
kubectl get secrets -n <ns>
kubectl describe secret <name> -n <ns>
kubectl get pod <pod> -n <ns> -o yaml
kubectl describe pod <pod> -n <ns>
kubectl logs <pod> -n <ns> (check volume / mount errors)
```

RESULT: Fix secret reference, permissions, or data.



6. RUNTIME INCIDENTS

Security incidents detected during runtime.

COMMON CAUSES

- Malicious container
- Privileged escalation
- Unexpected network
- File system changes
- Suspicious process

TROUBLESHOOTING STEPS

- 1 Detect and alert
- 2 Isolate affected workload
- 3 Collect evidence
- 4 Analyze root cause
- 5 Eradicate and patch
- 6 Monitor and validate

USEFUL COMMANDS

```
kubectl get pods -A
kubectl top pods -A
kubectl exec -it <pod> -- sh
ps aux
netstat -tulpn
kubectl logs <pod> -n <ns>
kubectl adm inspect <node>
```

RESULT: Contain the incident, remove threat, and restore securely.

7. PRODUCTION SECURITY TROUBLESHOOTING WORKFLOW



1. DETECT
Alert / Issue Identified



2. TRIAGE
Identify Scope and Impact



3. INVESTIGATE
Collect Logs, Events, Evidence



4. CONTAIN
Isolate / Block the Threat



5. RESOLVE
Fix Root Cause and Patch



6. VALIDATE
Test, Monitor, Verify Fix



7. DOCUMENT
Lessons Learned Update Playbooks

COMMON MISTAKES

- Guessing without evidence
- Skipping logs and events
- Not understanding the scope
- Making changes without backup
- Forgetting to validate the fix



VERIQTA PRODUCTION NOTE

Troubleshooting is a skill. Stay calm, follow the method, use the right tools, document everything, and make your systems stronger after every incident.



KEY TAKEAWAY

- Follow a process.
- Verify every step.
- Use the right commands.
- Document and learn.
- Secure systems. Protect users.



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36. ENTERPRISE KUBERNETES SECURITY ARCHITECTURE

SECURE BY DESIGN. PROTECT EVERY LAYER.
ZERO TRUST. DEFENSE IN DEPTH. CONTINUOUS ASSURANCE.

VERIQTA TIP

Security is not a product.
It is an architecture,
a process, and a culture.
Build security in every
layer from day one.

1. ZERO TRUST ARCHITECTURE



- ✓ Never trust, always verify
- ✓ Least privilege access
- ✓ Identity-based access
- ✓ Micro-segmentation
- ✓ Continuous verification
- ✓ Assume breach mindset
- ✓ Strong authentication
- ✓ Encrypt everywhere

2. DEFENSE IN DEPTH



- ✓ Multiple security layers
- ✓ Prevent, detect, respond
- ✓ Reduce blast radius
- ✓ Layered controls
- ✓ Redundancy and isolation
- ✓ Visibility at every layer
- ✓ Compensating controls
- ✓ Resilience by design

3. LAYERED SECURITY MODEL



- ✓ Perimeter security
- ✓ Identity & access security
- ✓ Network security
- ✓ Workload security
- ✓ Data security
- ✓ Runtime security
- ✓ Monitoring & detection
- ✓ Incident response

4. SECURE PLATFORM DESIGN



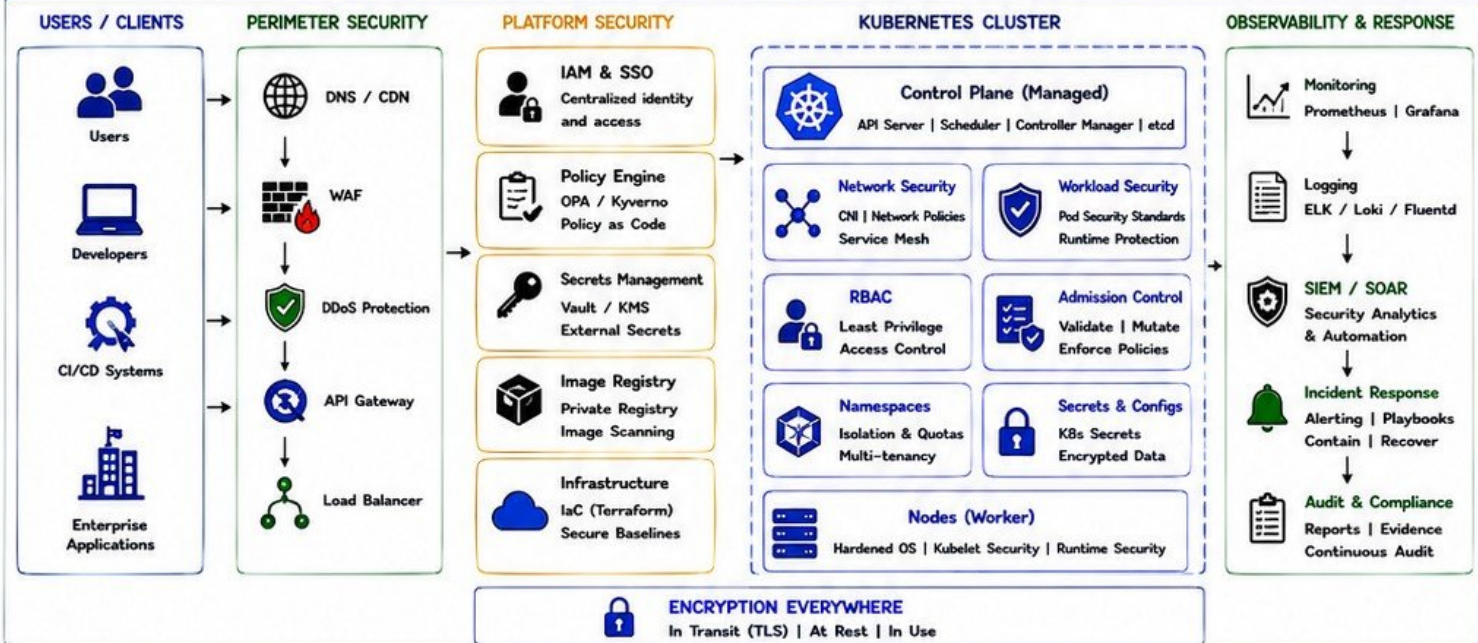
- ✓ Secure cluster build
- ✓ Automated compliance
- ✓ Policy as code
- ✓ Infrastructure as code
- ✓ GitOps for security
- ✓ Immutable infrastructure
- ✓ Secrets management
- ✓ Continuous hardening

5. REFERENCE ARCHITECTURE



- ✓ Enterprise ready
- ✓ Scalable and resilient
- ✓ Highly available
- ✓ Observable and auditable
- ✓ Governed and compliant
- ✓ Cloud agnostic
- ✓ Modular and extensible
- ✓ Future proof

6. ENTERPRISE KUBERNETES SECURITY REFERENCE ARCHITECTURE



7. SECURITY LAYERS (DEFENSE IN DEPTH)



COMMON RISKS

- ✗ Overly permissive access
- ✗ Exposed API servers
- ✗ Unscanned container images
- ✗ Weak secret management
- ✗ Missing network segmentation
- ✗ No monitoring or alerting

VERIQTA PRODUCTION NOTE



A secure Kubernetes platform is built with identity, policies, isolation, encryption, visibility, and automation. Design security as an architecture, not an afterthought.

KEY TAKEAWAY



- ✓ Adopt Zero Trust principles
- ✓ Implement Defense in Depth
- ✓ Secure every layer of the stack
- ✓ Automate security and compliance
- ✓ Continuously monitor and respond
- ✓ Security is everyone's responsibility



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VERIQTA TIP

Strong security is consistent execution of basics every day.
Discipline > tools.
Process > guesswork.

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37. PRODUCTION SECURITY BEST PRACTICES

SECURE TODAY. SECURE TOMORROW. SECURE ALWAYS.

1. CLUSTER HARDENING CHECKLIST



- ✓ Use least privilege RBAC
- ✓ Disable anonymous access
- ✓ Enable audit logging
- ✓ Enforce Network Policies
- ✓ Use Pod Security Standards (Restricted)
- ✓ Disable privileged containers
- ✓ Enable admission controllers
- ✓ Encrypt etcd with KMS
- ✓ Secure kubelet configuration
- ✓ Use read-only root filesystem
- ✓ Disable hostPath where not needed
- ✓ Rotate secrets regularly
- ✓ Enable image scanning
- ✓ Use minimal base images
- ✓ Enable runtime protection
- ✓ Review and remove unused components
- ✓ Enable MFA for access
- ✓ Regular vulnerability scans



GOAL:
Harden the cluster and reduce attack surface.

2. OPERATIONAL STANDARDS



- ✓ All changes via GitOps
- ✓ Infrastructure as Code
- ✓ Standardize namespaces
- ✓ Enforce resource quotas
- ✓ Use approved base images
- ✓ Secrets in external store (Vault / KMS)
- ✓ Centralized logging
- ✓ Centralized monitoring
- ✓ Alert on security events
- ✓ Validate deployments
- ✓ Automated policy checks
- ✓ Document runbooks
- ✓ Backup etcd regularly
- ✓ Test disaster recovery
- ✓ Access reviews quarterly
- ✓ Principle of least privilege
- ✓ Segregation of duties
- ✓ Change management process



GOAL:
Consistent, repeatable, and secure operations.

3. SECURITY REVIEWS



- ✓ Quarterly access reviews
- ✓ RBAC audit and cleanup
- ✓ Network Policy review
- ✓ Image vulnerability review
- ✓ Secrets and config review
- ✓ Admission policy review
- ✓ Audit log analysis
- ✓ Penetration testing
- ✓ Compliance gap analysis
- ✓ Cloud security posture review
- ✓ Threat modeling updates
- ✓ Third-party dependency scan
- ✓ Kubernetes version review
- ✓ Worker node security review
- ✓ Backup & restore review
- ✓ Incident readiness review
- ✓ Business continuity review
- ✓ Policy exception review



GOAL:
Continuously validate security posture.

4. UPGRADE STRATEGY



- ✓ Follow Kubernetes release schedule
- ✓ Test upgrades in staging environment
- ✓ Review release notes and CVEs
- ✓ Upgrade control plane first
- ✓ Upgrade worker nodes next
- ✓ Validate add-ons compatibility
- ✓ Use canary / phased upgrades
- ✓ Monitor after upgrade
- ✓ Have rollback plan ready
- ✓ Automate upgrade checks
- ✓ Document upgrade steps
- ✓ Communicate changes
- ✓ Regular dependency updates



GOAL:
Keep the platform secure and supported.

5. MAINTENANCE LIFECYCLE



- ✓ Daily: Monitor & alert
- ✓ Weekly: Patch scan
- ✓ Monthly: Vulnerability review
- ✓ Quarterly: Access review
- ✓ Quarterly: DR test
- ✓ Bi-annual: Pen test
- ✓ Annual: Security audit
- ✓ Update policies & docs
- ✓ Deprecate unused resources
- ✓ Rotate keys & secrets
- ✓ Review and tune alerts
- ✓ Capacity & performance review
- ✓ Review third-party integrations
- ✓ Automate compliance checks
- ✓ Continuously improve processes



GOAL:
Maintain security through the lifecycle.

6. KEY PRODUCTION SECURITY CONTROLS



IDENTITY

- Strong IAM / RBAC
- Least privilege
- MFA everywhere



NETWORK

- Network Policies
- Private clusters
- Encrypt in transit



WORKLOAD

- Pod Security Standards
- Image scanning
- Runtime protection



DATA

- Encrypt at rest
- Secrets management
- Backup protection



MONITORING

- Centralized logs
- Real-time alerts
- Threat detection



RESPONSE

- Incident playbooks
- Rapid containment
- Post-incident review

7. KUBERNETES SECURITY MAINTENANCE LIFECYCLE



ASSESS

- Identify risks
- Review posture
- Prioritize gaps



PLAN

- Define controls
- Set standards
- Build roadmap



IMPLEMENT

- Apply policies
- Automate security
- Harden platform



OPERATE

- Monitor & detect
- Respond to threats
- Maintain systems



REVIEW

- Measure & audit
- Learn & improve
- Update strategy



IMPROVE

- Refine controls
- Automate more
- Stay ahead

CONTINUOUS IMPROVEMENT

COMMON RISKS

- ✗ Over-permissioned access
- ✗ Unpatched vulnerabilities
- ✗ Exposed services
- ✗ Weak or missing network policies
- ✗ Secrets in code or images
- ✗ No monitoring or alerting
- ✗ No backups or DR testing



VERIQTA PRODUCTION NOTE

Security is not a one-time task. It is a continuous commitment. Build secure platforms, follow proven processes, and protect your users and data every day.



KEY TAKEAWAY

- ✓ Harden the cluster and workloads
- ✓ Standardize secure operations
- ✓ Review and validate regularly
- ✓ Keep everything updated
- ✓ Automate and monitor continuously
- ✓ Security is everyone's responsibility



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38. REAL PRODUCTION SECURITY CASE STUDIES

REAL INCIDENTS. REAL IMPACT. REAL LESSONS.

VERIQTA TIP

Every incident is a teacher. Learn, improve, and build stronger, more secure systems.

1. SUPPLY CHAIN COMPROMISE

WHAT HAPPENED
An attacker injected malicious code into a popular container image in a public registry. The image was automatically pulled by CI/CD pipelines and deployed to production.

IMPACT

- Malware executed in multiple clusters
- Data exfiltration and crypto-mining
- Service disruption and financial loss

ROOT CAUSE

- No image scanning or signature verification
- Trusted unsigned third-party images
- Lack of SBOM and provenance checks

LESSONS LEARNED

- Verify image signatures and provenance
- Scan images in pipeline and runtime
- Use vetted registries and pin versions
- Apply least privilege to workloads

2. EXPOSED DASHBOARD

WHAT HAPPENED
A Kubernetes dashboard was exposed on the internet without authentication due to a misconfigured Service and Ingress.

IMPACT

- Attackers gained cluster visibility
- Enumerated resources and workloads
- Attempted privilege escalation

ROOT CAUSE

- Dashboard exposed via LoadBalancer
- No authentication or IP restrictions
- Default dashboard used in production

LESSONS LEARNED

- Never expose dashboards publicly
- Use strong authentication (OIDC/OAuth)
- Restrict access with NetworkPolicies
- Remove unused or default components

3. PRIVILEGE ESCALATION

WHAT HAPPENED
A user with excessive permissions exploited a misconfigured RoleBinding to escalate privileges and create cluster admin access.

IMPACT

- Full cluster takeover by attacker
- Sensitive data exposure and tampering
- Backdoors created for persistence

ROOT CAUSE

- Overly permissive RBAC roles
- RoleBinding to broad user groups
- No regular access reviews

LESSONS LEARNED

- Apply least privilege RBAC
- Review and audit permissions regularly
- Use dedicated service accounts
- Enable RBAC access monitoring

4. CONTAINER ESCAPE

WHAT HAPPENED
A vulnerability in the container runtime allowed an attacker to escape the container and gain access to the underlying node.

IMPACT

- Node compromise
- Lateral movement to other workloads
- Cluster-wide risk

ROOT CAUSE

- Outdated container runtime
- HostPath mounts and privileged containers
- No runtime security controls

LESSONS LEARNED

- Keep runtimes and OS updated
- Avoid privileged containers
- Use Pod Security Standards / PSA
- Enable runtime threat detection

5. SECRET LEAKAGE

WHAT HAPPENED
Secrets were stored in ConfigMaps and logged in application debug logs, which were accessible to other users.

IMPACT

- Database credentials leaked
- Third-party API keys exposed
- Unauthorized access and data breach

ROOT CAUSE

- Secrets in ConfigMaps and env vars
- Secrets logged in plaintext
- No secret encryption at rest

LESSONS LEARNED

- Use Kubernetes Secrets or external vaults
- Encrypt secrets at rest and in transit
- Avoid logging sensitive data
- Rotate secrets regularly

6. MISCONFIGURED NETWORK POLICY

WHAT HAPPENED
Default allow-all network policy allowed pods to communicate with sensitive internal services and external endpoints.

IMPACT

- Lateral movement within cluster
- Data exfiltration to external servers
- Compliance and regulatory violations

ROOT CAUSE

- No default deny policies
- Missing egress restrictions
- Lack of network segmentation

LESSONS LEARNED

- Implement default deny NetworkPolicies
- Restrict egress to required endpoints
- Segment workloads by sensitivity
- Continuously test network controls

7. KEY TAKEAWAYS FROM REAL INCIDENTS



SECURE THE SUPPLY CHAIN
Verify, scan, and trust only what you absolutely need.



LIMIT EXPOSURE
Expose nothing unnecessary. The internet is hostile.



LEAST PRIVILEGE
Access should be just enough, just-in-time.



HARDEN RUNTIME
Secure nodes, runtimes, and workloads continuously.



PROTECT SECRETS
Treat secrets like crown jewels. Store, encrypt, and rotate.



SEGMENT & MONITOR
Segment networks and monitor everything continuously.

AVOID THESE MISTAKES

- Using default configurations
- Over-permissioned access
- Exposing internal services
- Storing secrets insecurely
- Skipping updates and scans
- No monitoring or alerting



VERIQTA PRODUCTION NOTE

Security is a continuous journey, not a destination. Learn from incidents, strengthen your defenses, and build a culture of security in every deployment.



FINAL REMINDER

Prepare before an incident. Detect early. Respond quickly. Recover fully. Learn always. Make the next incident impossible.



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39. SENIOR KUBERNETES SECURITY REVIEW

REVIEW DEEPLY. THINK CRITICALLY. SECURE CONFIDENTLY.
SECURITY IS ARCHITECTURE. ARCHITECTURE IS SECURITY.

VERIQTA TIP

Great security reviews focus on risk, context, and impact — not just checklists.

1. ARCHITECTURE REVIEW METHODOLOGY



DEFINE CONTEXT

Understand business objectives, compliance requirements, workloads, data sensitivity, and threat landscape.



COLLECT INFORMATION

Gather architecture diagrams, cluster configs, IAM roles, policies, network design, and operational procedures.



IDENTIFY & ANALYZE RISKS

Map assets, trust boundaries, and data flows. Identify threats and evaluate existing security controls.



EVALUATE CONTROLS

Assess control effectiveness, configuration correctness, compensating controls, and defense in depth.



RECOMMEND & PRIORITIZE

Provide actionable recommendations with severity, effort, impact, and remediation roadmap.

2. SECURITY ASSESSMENT WORKFLOW



1. PLAN

- Define scope
- Identify systems
- Set objectives
- Stakeholders



2. DISCOVER

- Cluster discovery
- Asset inventory
- Data classification
- Trust boundaries



3. ANALYZE

- Threat modeling
- Risk assessment
- Control mapping
- Gap analysis



4. EVALUATE

- Control testing
- Config review
- Policy review
- Pen testing (if applicable)



5. REPORT

- Findings summary
- Risk ratings
- Evidence
- Recommendations



6. REMEDIATE

- Fix issues
- Improve controls
- Update policies
- Validate fixes



7. REVIEW

- Re-assess
- Measure maturity
- Continuous improvement

3. COMMON ENTERPRISE FINDINGS

- ✗ Overly permissive RBAC and cluster-admin sprawl
- ✗ ServiceAccounts with excessive permissions
- ✗ Default namespaces and weak isolation
- ✗ Unaudited third-party containers and images
- ✗ Secrets stored in plaintext or code repositories
- ✗ Missing NetworkPolicies or overly open rules
- ✗ Exposed dashboards and endpoints
- ✗ No image scanning or vulnerability management
- ✗ Lack of logging, monitoring, and alerting
- ✗ Outdated Kubernetes versions and components
- ✗ Inadequate backup, DR, and etcd protection
- ✗ No admission controls or policy enforcement
- ✗ Insufficient segmentation between environments
- ✗ Weak key management and encryption posture

4. KUBERNETES SECURITY MATURITY MODEL

LEVEL 0 AD HOC

Minimal security controls. Reactive. High risk of breaches.

LEVEL 1 FOUNDATIONAL

Basic controls implemented. Some policies and visibility. Inconsistent.

LEVEL 2 DEFINED

Standardized policies, RBAC, logging, and monitoring in place.

LEVEL 3 MANAGED

Automated enforcement, threat detection, and continuous monitoring.

LEVEL 4 OPTIMIZED

Proactive security, zero trust, advanced detection, and continuous improvement.



GOAL: Continuously improve maturity and reduce risk over time.

5. DESIGN RECOMMENDATIONS

- ✓ Adopt Zero Trust architecture across the cluster
- ✓ Enforce least privilege RBAC and Just-In-Time access
- ✓ Use NetworkPolicies for all namespace communications
- ✓ Implement admission control with OPA / Kyverno
- ✓ Scan images and enforce trusted registries
- ✓ Encrypt everything: in transit and at rest
- ✓ Centralize secrets with external vaults (e.g., Vault, AWS Secrets Manager)
- ✓ Enable full observability: logs, metrics, traces
- ✓ Automate security checks in CI/CD pipelines
- ✓ Segment environments and workloads
- ✓ Harden nodes, disable unnecessary services
- ✓ Regularly patch and upgrade Kubernetes
- ✓ Test backups and disaster recovery regularly
- ✓ Continuously assess, monitor, and improve

6. KUBERNETES SECURITY ASSESSMENT DOMAINS



IDENTITY & ACCESS

- IAM Integration
- RBAC & Least Privilege
- ServiceAccounts
- MFA / SSO



NETWORK SECURITY

- NetworkPolicies
- Ingress Security
- CNI Security
- Encryption



WORKLOAD SECURITY

- Pod Security Standards
- Image Security
- Runtime Security
- Vulnerability Mgmt



DATA PROTECTION

- Secrets Management
- Encryption at Rest
- Data Classification
- Backup Security



PLATFORM SECURITY

- Cluster Hardening
- Kubelet Security
- API Server Security
- etcd Security



MONITORING & DETECTION

- Logging & Auditing
- Threat Detection
- Alerting
- SIEM Integration



GOVERNANCE & COMPLIANCE

- Policies & Standards
- Compliance Mapping
- Audit & Evidence
- Risk Management

COMMON REVIEW MISTAKES

- ✗ Focusing only on configurations, not risks
- ✗ Ignoring business context and data sensitivity
- ✗ No validation of controls in practice
- ✗ Overlooking third-party integrations
- ✗ Skipping runtime and workload analysis
- ✗ Weak follow-up and remediation tracking



VERIQTA PRODUCTION NOTE

A senior security review is not a one-time event. It is a continuous function that blends architecture, operations, and security engineering to protect the business, reduce risk, and build trust.

KEY TAKEAWAY



- ✓ Review with a risk-first mindset.
- ✓ Validate, don't just review documents.
- ✓ Defense in depth is not optional.
- ✓ Automate, monitor, and improve.
- ✓ Security is a continuous journey.



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VERIQTA TIP

A secure Kubernetes platform is built by design, validated by controls, and proven in production.

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40. PRODUCTION READINESS CHECKLIST

SECURE BY DEFAULT. OPERATE WITH CONFIDENCE.

NOT CHECKED? NOT READY.



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1. CONTROL PLANE SECURITY



- ☐ Control plane in private network
- ☐ API server behind firewall / ALB
- ☐ TLS 1.2+ for all components
- ☐ mTLS enabled for component communication
- ☐ API server audit logging enabled
- ☐ Encryption at rest for etcd
- ☐ etcd access restricted to control plane only
- ☐ Regular control plane patching
- ☐ Kubernetes version supported
- ☐ Backup of etcd tested and verified

2. NODE SECURITY



- ☐ Nodes in private subnets
- ☐ Host firewall enabled
- ☐ Only required ports open
- ☐ kubelet authentication enabled
- ☐ Read-only root filesystem
- ☐ Swap disabled
- ☐ Automatic OS patching
- ☐ CIS benchmark applied
- ☐ Container runtime hardened
- ☐ Unnecessary services disabled
- ☐ Access via SSH key + bastion
- ☐ Node labels and taints enforced

3. RBAC REVIEW



- ☐ RBAC least privilege enforced
- ☐ No wildcard permissions
- ☐ ClusterRoles reviewed
- ☐ RoleBindings reviewed
- ☐ ServiceAccounts scoped
- ☐ No default ServiceAccount usage in workloads
- ☐ Privileged access minimized
- ☐ Access review performed
- ☐ Inactive accounts removed
- ☐ Periodic RBAC audit enabled

4. NETWORK SECURITY



- ☐ NetworkPolicies applied
- ☐ Default deny ingress & egress
- ☐ Only required traffic allowed
- ☐ Pod-to-Pod isolation enabled
- ☐ Ingress controller secured
- ☐ TLS for all external traffic
- ☐ DDoS protection enabled
- ☐ Private cluster endpoints
- ☐ DNS policies restricted
- ☐ CNI plugin security enabled
- ☐ VPC / subnet segmentation
- ☐ Network flow monitoring on

5. SECRETS MANAGEMENT



- ☐ Secrets encrypted at rest
- ☐ Use external secrets / vault
- ☐ No hardcoded secrets
- ☐ Secrets rotation enabled
- ☐ Secrets access via RBAC
- ☐ Least privilege for secret access
- ☐ KMS or Vault integration
- ☐ SealedSecrets / ExternalSecrets
- ☐ Secrets not logged or exposed
- ☐ Backup of secrets store tested

6. RUNTIME PROTECTION



- ☐ Pod Security Standards enforced
- ☐ No privileged containers
- ☐ Drop all Linux capabilities
- ☐ Read-only root filesystem
- ☐ AppArmor / Seccomp enabled
- ☐ Only trusted images allowed
- ☐ Runtime threat detection on
- ☐ Falco or equivalent enabled
- ☐ Exec into containers restricted
- ☐ Resource limits on all workloads

7. MONITORING



- ☐ Cluster metrics collection on
- ☐ Node and pod metrics on
- ☐ Kubernetes component metrics
- ☐ Alert rules configured
- ☐ SLOs / SLIs defined
- ☐ Dashboards for key signals
- ☐ Anomaly detection enabled
- ☐ Capacity monitoring enabled
- ☐ Certificate / expiry monitoring
- ☐ Third-party integrations on
- ☐ On-call alerting tested

8. LOGGING



- ☐ Centralized logging enabled
- ☐ Audit logs collected
- ☐ Application logs collected
- ☐ System logs collected
- ☐ Logs retained per policy
- ☐ Logs immutable / tamper proof
- ☐ Log access restricted
- ☐ Log search and alerts enabled
- ☐ Sensitive data redacted
- ☐ Log backup and restore tested
- ☐ Compliance log requirements met

9. COMPLIANCE



- ☐ Security policies documented
- ☐ Access reviews scheduled
- ☐ Baseline compliance standard
- ☐ CIS Kubernetes benchmark
- ☐ Compliance reports generated
- ☐ Vulnerability management
- ☐ Third-party risk assessed
- ☐ Data classification applied
- ☐ Regulatory requirements met
- ☐ Evidence collection automated
- ☐ External audit ready

10. INCIDENT RESPONSE



- ☐ Incident response plan defined
- ☐ Roles and responsibilities clear
- ☐ Communication channels ready
- ☐ Playbooks documented
- ☐ Security runbooks available
- ☐ Tools and access validated
- ☐ Forensics & log retention ready
- ☐ Backup & restore procedures tested
- ☐ Post-incident review process
- ☐ Tabletop exercises performed
- ☐ Lessons learned tracked

11. FINAL ENTERPRISE SECURITY PRINCIPLES

- | | |
|------------------------|---|
| ✓ SECURITY BY DESIGN | Build security into architecture, not as an afterthought. |
| ✓ ZERO TRUST | Verify every request. Trust nothing. Always authenticate and authorize. |
| ✓ LEAST PRIVILEGE | Give minimum access needed. Nothing more. |
| ✓ DEFENSE IN DEPTH | Layer controls. Assume breach. Reduce blast radius. |
| ✓ AUTOMATE SECURITY | Automate enforcement, scanning, patching, and compliance. |
| ✓ OBSERVE CONTINUOUSLY | Monitor everything. Detect early. Respond fast. |
| ✓ IMPROVE CONTINUOUSLY | Learn from incidents. Test. Measure. Improve. |

PRODUCTION READINESS SCORECARD



90 - 100	✓ EXCELLENT	PRODUCTION READY.
70 - 89	▲ GOOD	MINOR GAPS
50 - 69	▲ FAIR	IMPROVEMENTS NEEDED
0 - 49	✗ POOR	NOT READY.



VERIQTA PRODUCTION NOTE

A production-ready Kubernetes platform is secure, resilient, observable, and compliant. Validate. Verify. Continuously improve. Security is a process, not a product.

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